



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

“Electronics and Communication Engineering”

III to VIII Semester

Jun/Jul-2024

2018 & 2022 Scheme

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Third Semester B.E. Degree Examination, June/July 2024 Network Theory

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive the expression for,
 - (i) Δ to Y transformation
 - (ii) Y to Δ transformation

(10 Marks)
- b. Determine the equivalent resistance between A and B of the network shown in Fig. Q1 (b).
(10 Marks)

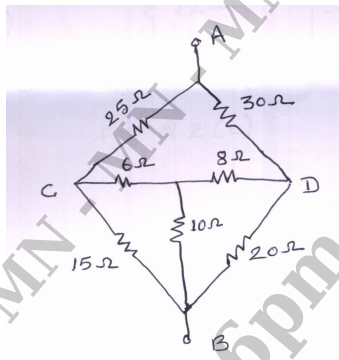


Fig. Q1 (b)

OR

- 2 a. Determine the current i_2 and voltage v_1 for the circuit shown in Fig. Q2 (a).
(10 Marks)

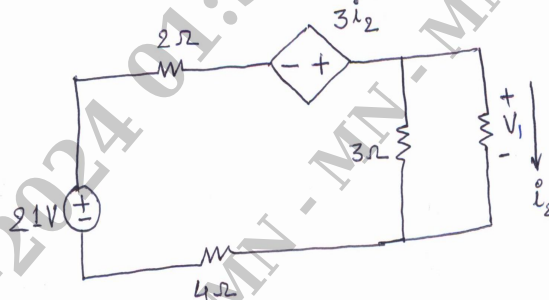


Fig. Q2 (a)

- b. Determine the value of V_2 , such that current through 4Ω resistor is zero, using mesh current analysis method for the network shown in Fig. Q2 (b).
(10 Marks)

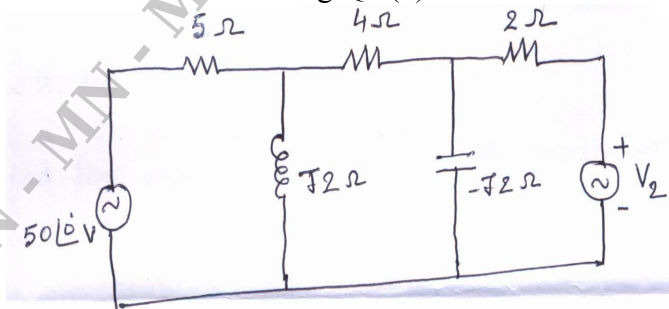


Fig. Q2 (b)

Module-2

- 3 a. State Super position theorem. Using superposition theorem, find the voltage V_1 across 3Ω resistor for the Network shown in Fig. Q3 (a). (10 Marks)

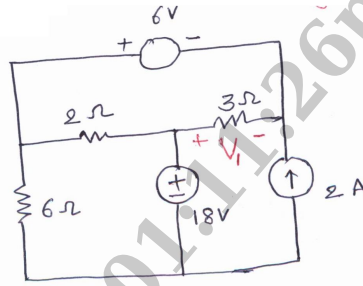


Fig. Q3 (a)

- b. Evaluate the current through the load resistor R_L for the circuit shown in Fig. Q3 (b) using Millman's Theorem. (10 Marks)

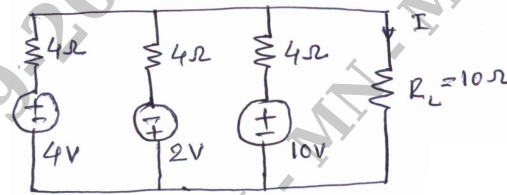


Fig. Q3 (b)

OR

- 4 a. Explain the procedure to find Norton's equivalent resistance in a network which has both dependent and independent sources with an example. (06 Marks)
- b. Find the value of Z_L for which maximum power transfer occurs in the circuit shown in Fig. Q4 (b). (04 Marks)

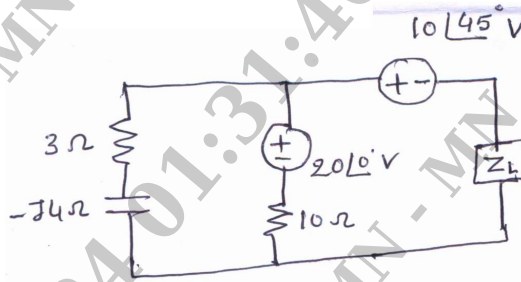


Fig. Q4 (b)

- c. Determine the current flowing through the 6Ω resistor for the circuit shown in Fig. Q4 (c) using Thevenin's theorem. (10 Marks)

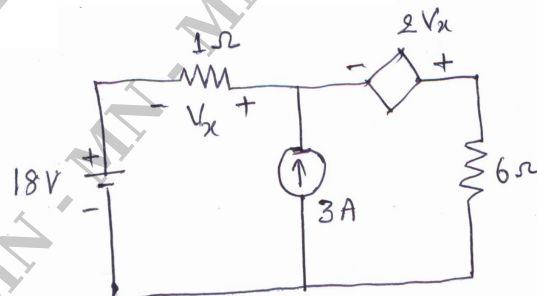


Fig. Q4 (c)

Module-3

- 5 a. Explain the transient behavior of R, L and C. Also explain the procedure for evaluating transient behavior. (10 Marks)

- b. In the circuit shown in Fig. Q5 (b) the switch 'S' is moved from a to b at $t = 0$. Evaluate the values of i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t = 0^+$. If $R = 1 \Omega$, $L = 1 \text{ H}$, $C = 0.1 \mu\text{F}$ and $V = 100 \text{ V}$. Assume steady state is achieved when K is at 'a'.

(10 Marks)

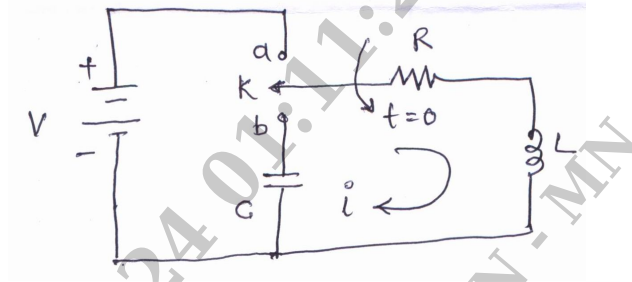


Fig. Q5 (b)

OR

- 6 a. Evaluate i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t = 0^+$ for the circuit shown in Fig. Q6 (a), when switch K is changed from position 1 to 2 at $t = 0$, the steady state having been reached before switching.

(10 Marks)

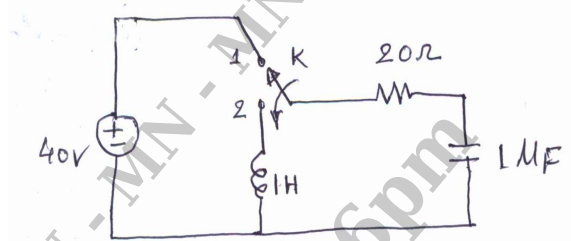


Fig. Q6 (a)

- b. Find the values of i_1, i_2 , $\frac{di_1}{dt}$, $\frac{di_2}{dt}$, $\frac{d^2i_1}{dt^2}$ and $\frac{d^2i_2}{dt^2}$ at $t = 0^+$ for the circuit shown in Fig. Q6 (b).

(10 Marks)

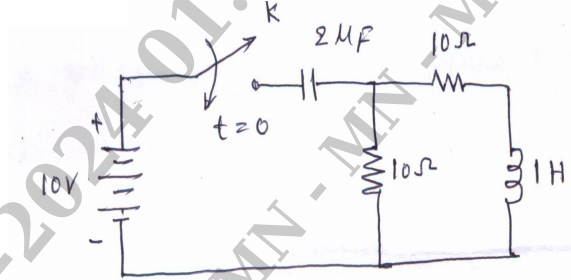


Fig. Q6 (b)

Module-4

- 7 a. Obtain Laplace transform of,
 (i) Step function
 (ii) Ramp function
 (iii) Impulse function.

(10 Marks)

- b. Find the Laplace transform of the periodic waveform shown in Fig. Q7 (b).

(10 Marks)

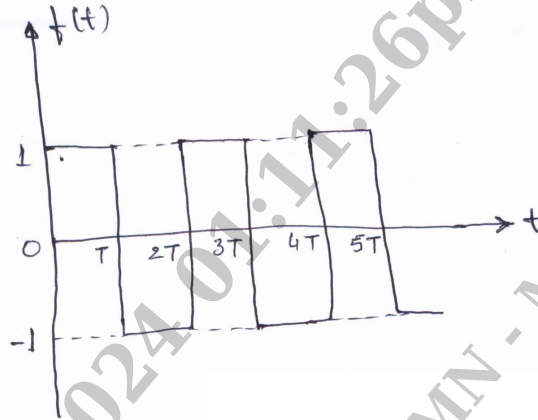


Fig. Q7 (b)

OR

- 8 a. Deduce the Laplace transform of the following :

- (i) $\sin^2 t$
- (ii) $\cos^2 t$
- (iii) $\sin \omega t$

(10 Marks)

- b. State and prove Initial and Final value theorems.

(10 Marks)

Module-5

- 9 a. Express Z-parameters in terms of h-parameters and what are hybrid parameters.
b. Determine the transmission parameters for the network shown in Fig. Q9 (b).

(10 Marks)

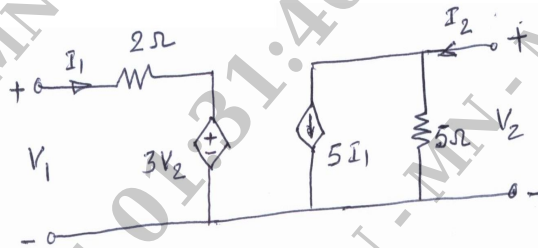


Fig. Q9 (b)

(10 Marks)

OR

- 10 a. Define the followings :

- (i) Resonance
- (ii) Q-factor
- (iii) Band width
- (iv) Selectivity.

(08 Marks)

- b. Prove that the resonating frequency in a R-L-C series circuit is geometrical mean of half power frequencies is $f_0 = \sqrt{f_1 f_2}$.

(12 Marks)

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Third Semester B.E. Degree Examination, June/July 2024

Digital System Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Design a combinational logic circuit so that an output is generated indicating when a majority of four inputs is true. (06 Marks)
- b. Place the following equations into the proper canonical form
- i) $f(w, x, y, z) = \bar{w}x + y\bar{z}$ ii) $f(A, B, C, D) = A + \bar{B} + C)(\bar{A} + D)$ (06 Marks)
- c. Using K-map determine minimal sum of product expressions and implement the simplified equation using only NAND gates $f(w, x, y, z) = \sum m(1, 2, 3, 4, 9) + \sum d(10, 11, 12, 13, 14, 15)$ (08 Marks)

OR

- 2 a. Define the following terms literal, canonical sum of products, Karnaugh Map, Prime implicants. (04 Marks)
- b. Find the minimal sum of the following Boolean function using Quine McClusky method $f(w, x, y, z) = \sum(1, 3, 13, 15) + \sum d(8, 9, 10, 11)$ (08 Marks)
- c. Using K-map determine minimal product of sum expression and implement the simplified equation using only NOR gates $f(a, b, c, d) = \pi M(0, 4, 5, 7, 8, 9, 11, 12, 13, 15)$. (08 Marks)

Module-2

- 3 a. Implement following multiple output function using 74LS138 decoder
 $F_1(A, B, C) = \sum m(1, 4, 5, 7)$
 $F_2(A, B, C) = \pi m(2, 3, 6, 7)$ (06 Marks)
- b. Explain 4-bit carry look ahead adder with necessary diagram and relevant expression. (10 Marks)
- c. Implement $f(a, b, c, d) = \sum m(0, 1, 5, 6, 7, 9, 10, 15)$ using 8 : 1 MUX with a, b, c as select lines (04 Marks)

OR

- 4 a. Implement full adder using 74138 decoder. (06 Marks)
- b. Design a 2-bit Magnitude comparator. (08 Marks)
- c. Design 4-line to 2 line priority uncoder which gives MSB the highest priority and LSB least priority. (06 Marks)

Module-3

- 5 a. What is race around condition? Explain JK master slave flip-flop with diagram function table and timing diagram. (08 Marks)
- b. Explain the working of 4-bit Johnson counter using necessary diagram and waveform. (06 Marks)
- c. Explain with a neat diagram and truth table, a 4-bit SIPO shift register to store binary number 1010. (06 Marks)

OR

- 6 a. Explain the operation of switch debouncer using SR latch with the help of circuit and waveform. (06 Marks)
 b. Explain the working of 3-bit Asynchronous up-down counter with necessary waveform and truth table. (10 Marks)
 c. Write the difference between combinational circuits and sequential circuits. (04 Marks)

Module-4

- 7 a. Design a synchronous Mod -6 counter using clocked D- Flip-Flop. (10 Marks)
 b. Design a Moore type sequence detector to detect a serial input sequence of 101. (10 Marks)

OR

- 8 a. Construct the excitation table, transition table and state diagram for the sequential circuit shown in Fig Q8(a).

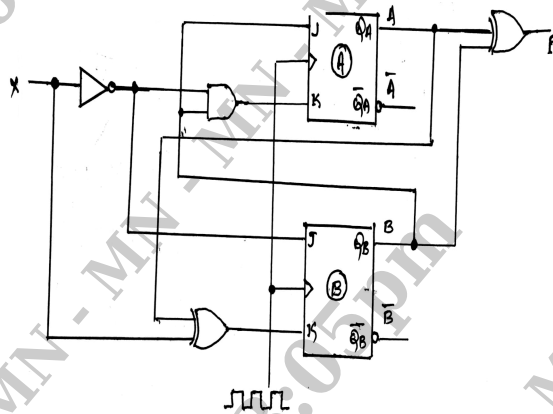


Fig Q8(a)

- b. Design a synchronous decade counter using T-flip flop and draw the logic diagram. (10 Marks)

Module-5

- 9 a. List the guidelines for construction of state graphs. (10 Marks)
 b. Design a sequential circuit to convert BCD to excess - 3 code with state table state graph and transition table. (10 Marks)

OR

- 10 a. Explain with block diagram design of serial Adder with accumulator. (10 Marks)
 b. Explain with block diagram design of Binary multiplier. (10 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024 Analog Circuits

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. Explain the design concept of common emitter collector to Base feedback resistor biasing circuit and explain how collector to base feedback resistor provides a negative feedback in the circuit. (07 Marks)
 - b. Considering the conceptual circuit of common source MOSFET amplifier, derive the expression for transconductance g_m and voltage gain A_v . (08 Marks)
 - c. For common emitter voltage divider circuit having $\beta = 100$, $R_1 = 10 \text{ K}\Omega$, $R_2 = 5 \text{ K}\Omega$, $R_C = 1 \text{ K}\Omega$ and $R_E = 500 \Omega$ is provided with DC biasing voltage $V_{CC} = 10 \text{ V}$, Calculate V_{CE} and I_C . (05 Marks)

OR

- 2
 - a. Derive an expression for small signal collector current, transconductance g_m and voltage gain A_v in BJT, when small signal V_{bc} is applied between base and emitter. (10 Marks)
 - b. Design voltage divider bias circuit using MOSFET to establish $I_D = 0.5 \text{ mA}$ and MOSFET parameter are $V_t = 1 \text{ V}$ and $K'_n \left(\frac{W}{L} \right) = 0.5 \text{ mA/V}^2$. Assume $V_{DD} = 15 \text{ V}$. (10 Marks)

Module-2

- 3
 - a. Explain Three basic configurations of MOSFET amplifier and derive expression for characteristic parameter of amplifiers. (08 Marks)
 - b. Briefly explain the Barkhausen criteria for oscillation. (04 Marks)
 - c. For an n-channel MOSFET with $t_{ox} = 10 \text{ nm}$, $L = 1 \mu\text{m}$, $W = 10 \mu\text{m}$, $L_{ov} = 0.05 \mu\text{m}$, $C_{Sbo} = C_{dbo} = 10 \text{ fF}$, $V_O = 0.6 \text{ V}$, $V_{SB} = 1 \text{ V}$, $V_{DS} = 2 \text{ V}$. Calculate the following capacitance when the transistor is operating in saturation,
 (i) C_{OX} (ii) C_{OV} (iii) C_{gs} (iv) C_{gd} (v) C_{sb} and C_{db} .
 Consider $\epsilon_{ox} = 3.45 \times 10^{-11}$ (08 Marks)

OR

- 4
 - a. Explain the working of RC phase shift oscillator and show how RC network provides 180° of phase shift. (08 Marks)
 - b. In a transistor Calpitts oscillator $C_1 = 1 \text{ nF}$ and $C_2 = 1000 \text{ nF}$. Find the value of L for a frequency of 100 kHz . (04 Marks)
 - c. Explain the High frequency response of common source MOSFET amplifier with its equivalent circuit. (08 Marks)

Module-3

- 5 a. Explain the effect of negative feedback on input and output resistance of voltage series feedback amplifier. (10 Marks)
- b. Explain transformer coupled Class A power amplifier and show that the maximum efficiency of transformer coupled Class A power amplifier is 50%. (10 Marks)

OR

- 6 a. Draw the block diagram of four types of feedback topologies and compare them with respect to input and output resistance. (10 Marks)
- b. Compare Class B pushpull and complementary symmetry power amplifiers. (04 Marks)
- c. In a Class B push pull amplifier operating with $V_{CC} = 25V$ provides a 22 V peak signal to an $8\ \Omega$ load. Find
- (i) Peak load current (ii) dc current drawn from the supply (iii) input power
- (iv) Output current efficiency (v) power dissipation (06 Marks)

Module-4

- 7 a. State the ideal op-amp characteristics. (06 Marks)
- b. Design a linear combination circuit using op-amp to obtain output $V_0 = -2V_1 - 8V_2 - V_3$ with $R_{in} \geq 20k\Omega$ at all the inputs and all the resistances $\leq 200k\Omega$ (04 Marks)
- c. Draw the circuit of 3 op-amp instrumentation amplifier and derive the expression for its output voltage. (10 Marks)

OR

- 8 a. Explain the working of voltage follower using op-amp and show that its gain is unity. State its advantages. (06 Marks)
- b. Explain the working of zero crossing detectors. (06 Marks)
- c. Design an inverting Schmitt trigger to have trigger voltages of $\pm 4V$ using op-amp 741 with supply of $\pm 15V$. Consider $I_{B(max)} = 500\text{ nA}$. (08 Marks)

Module-5

- 9 a. With neat circuit diagram, explain the operation of R-2R D/A converter. (10 Marks)
- b. Explain the working of pulse width modulation circuit using 555 IC. (06 Marks)
- c. Design a low pass filter using op-amp at a cut off frequency of 1 kHz with pass gain of 2 and choose $C = 0.01\ \mu F$ (04 Marks)

OR

- 10 a. Explain with neat circuit diagram the working of positive precision Half Wave Rectifier. (06 Marks)
- b. Design a monostable 555 timer circuit to produce an output pulse of 10 sec wide and draw the circuit diagram. Choose $C = 100\ \mu F$. (04 Marks)
- c. Draw the circuit of second order low pass filter and explain its operation. (10 Marks)

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18EC43

Fourth Semester B.E. Degree Examination, June/July 2024 Control Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define closed loop control system. Compare open loop and closed loop control system. (06 Marks)
- b. Write the differential equations of performance for the mechanical system in Fig.Q1(b). Draw its F – V analogous circuit.

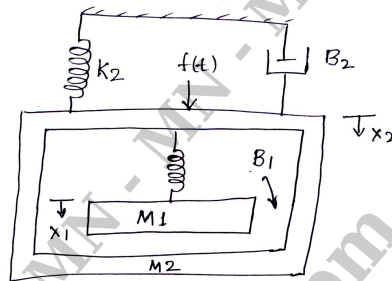


Fig.Q1(b)

(14 Marks)

OR

- 2 a. What are the advantages of using negative feedback in control system? (06 Marks)
- b. Draw the F – V and F – I analogues circuits for the mechanical system shown in Fig.Q2(b) with necessary equations.

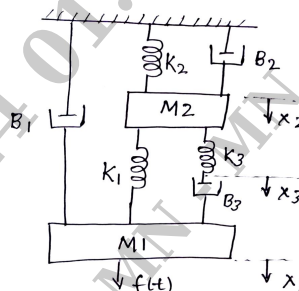


Fig.Q2(b)

(14 Marks)

Module-2

- 3 a. State advantages and disadvantages of the block diagram reduction technique. (06 Marks)
- b. For the block diagram shown in Fig.Q3(b), determine the transfer function : $\frac{Q_2(s)}{Q(s)}$ using block diagram reduction algebra.

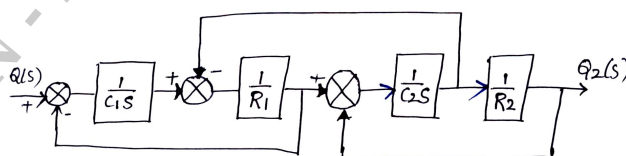


Fig.Q3(b)

(10 Marks)

- c. Derive the transfer function of simple closed loop system. (04 Marks)

OR

- 4 a. Find $\frac{C}{R}$ using Mason's gain formula for the signal flow graph shown in Fig.Q4(a).

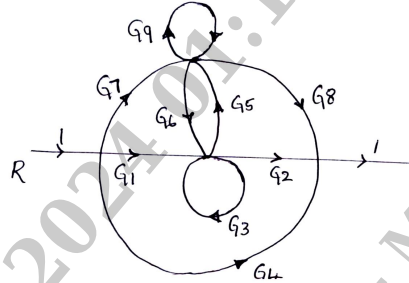


Fig.Q4(a)

(10 Marks)

- b. For the block diagram given in Fig.Q4(b) obtain overall transfer function using Mason's gain formula.

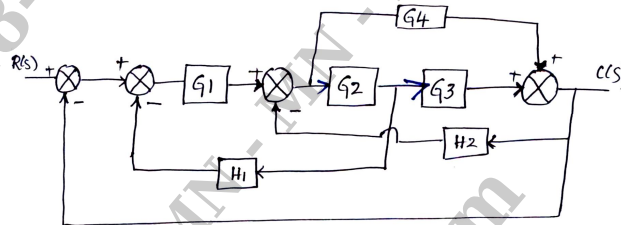


Fig.Q4(b)

(10 Marks)

Module-3

- 5 a. With the help of graphical representation and mathematical expressions, explain the following signals :
- Step signal
 - Ramp signal
 - Impulse signal
 - Parabolic signal.
- b. Derive an expression for the under-damped response of a second order feedback control system for step input.

(10 Marks)

(10 Marks)

OR

- 6 a. For the system shown in the Fig.Q6(a)
- Identify the type of $\frac{C(s)}{E(s)}$
 - Find the values of k_p , k_v , k_a
 - If $r(t) = 10u(t)$, find the steady state value of the output.

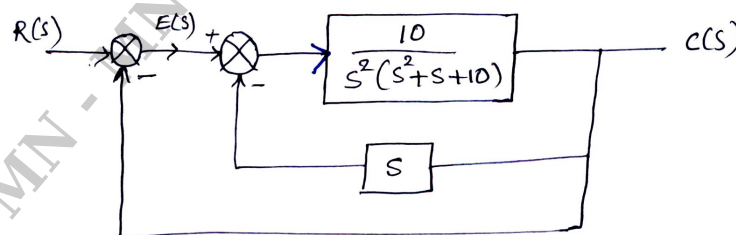


Fig.Q6(a)

(10 Marks)

- b. Discuss the various types of controllers used in the control systems.

(10 Marks)

Module-4

- 7 a. $S^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$. Find the number of roots of this equation with positive real part, zero real part and negative real part using Routh's criterion. (10 Marks)
- b. Find the gain margin and phase margin analytically for the negative feedback control system having open loop T.F. $G(s)H(s) = \frac{6}{(s^2 + 2s + 2)(s + 2)}$. (10 Marks)

OR

- 8 a. The open loop transfer function of a control system is given by $G(s) = \frac{K}{s(s+2)(s^2+6s+25)}$. Sketch the complete root-locus as K is varied from 0 to infinity. (15 Marks)
- b. Explain experimental determination of frequency response. (05 Marks)

Module-5

- 9 a. For a feedback control system : $G(s)H(s) = \frac{40}{(s+4)(s^2+2s+2)}$. Find gain margin and stability from Nyquist plot. (10 Marks)
- b. Draw polar plot of $G(s)H(s) = \frac{100}{(s+2)(s+4)(s+8)}$. (10 Marks)

OR

- 10 a. Obtain the state equation and output equation of the electric networks as shown in Fig.Q10(a).

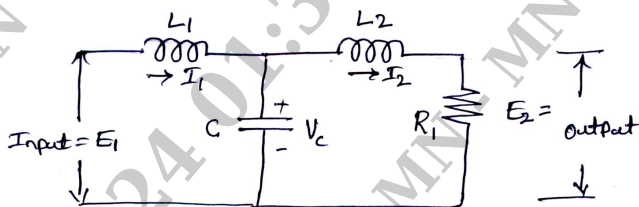


Fig.Q10(a)

- b. State the effect of lead, lag and lead-lag compensating networks. (10 Marks)

Fifth Semester B.E. Degree Examination, June/July 2024

Digital Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Determine the 6-DFT of the data sequence $x(n) = \{1, 1, 2, 2, 3, 3\}$ and Compute the corresponding amplitude and phase spectrum. (10 Marks)
- b. State and prove the following properties of DFT's :
 - (i) Linearity property
 - (ii) Periodicity property (10 Marks)

OR

- 2 a. Find the N-point DFT of $x(n) = \cos\left(\frac{2\pi}{N} K_0 n\right)$; $0 \leq n \leq N-1$. (05 Marks)
- b. Find the 4-point DFT of the sequence $x(n) = \{1, 2, 0, 1\}$ using matrix method. (05 Marks)
- c. Find the circular convolution of given data sequence $x_1(n) = \{1, 3, 5, 7\}$ and $x_2(n) = \{2, 4, 6, 8\}$, using DFT-IDFT method. (10 Marks)

Module-2

- 3 a. Determine the output sequence of a FIR filter whose impulse response is $h(n) = \{1, 1, 1\}$ and input sequence $x(n) = \{3, -1, 0, 1, 3, 2, 0, 1, 2, 1\}$ using overlap-add method. Assume length of block is 6. (10 Marks)
- b. Determine 8-point DFT Sequence for given input signal $x(n) = n+1$ using DIF-FFT algorithm. (10 Marks)

OR

- 4 a. Find the response of LTI system (Linear convolution) of input sequence $x(n) = \{1, 1, 1\}$ and impulse response $h(n) = \{-1, -1\}$ using DIT-FFT algorithm. (12 Marks)
- b. What is total number of complex additions and multiplications required to compute $N = 1024$ point DFT using direct and FFT method and also calculate the percentage savings in multiplications and additions. (08 Marks)

Module-3

- 5 a. With necessary mathematical analysis, explain the frequency sampling technique of FIR filter design. (10 Marks)
- b. The desired frequency response of a low pass filter is given by,

$$H_d(w) = \begin{cases} e^{-j3w}; & |w| \leq \frac{3\pi}{4} \\ 0 & ; \frac{3\pi}{4} < |w| < \pi \end{cases}$$

Determine the frequency response of the FIR filter if Hamming window is used. (10 Marks)

OR

- 6 a. List the steps in the design of a FIR filter using window functions. (05 Marks)
 b. Realize the Linear FIR filter having the following transfer function,
 $H(z) = 1 + 0.25z^{-1} - 0.125z^{-2} + 0.25z^{-3} + z^{-4}$ (05 Marks)
 c. Sketch the lattice realization for given FIR filter with the following difference equation,
 $y(n) = x(n) + 3.1x(n-1) + 5.5x(n-2) + 4.2x(n-3) + 2.3x(n-4)$ (10 Marks)

Module-4

- 7 a. Derive an expression for order and cut off frequency of a low pass Butterworth filter. (08 Marks)
 b. Design a butterworth digital low pass filter with maximum pass band attenuation of 3 db at 500 Hz, minimum attenuation of 15 db at stopband edge frequency of 750 Hz and sampling frequency $F_s = 2$ KHz. Use bilinear transformation method. (Assume $T = 1$ sec) (12 Marks)

OR

- 8 a. Derive mapping function used in transforming analog filter to digital filter by bilinear transformation. (08 Marks)
 b. Distinguish between FIR and IIR filters. (04 Marks)
 c. Obtain the direct form I and direct form II realization for the following system :
 $y(n) + 0.1y(n-1) - 0.2y(n-2) + 3x(n) + 3.6x(n-2) + 0.6x(n-2)$ (08 Marks)

Module-5

- 9 a. With neat diagrams, explain hardware units used in DSP processors. (10 Marks)
 b. Find the signed Q-15 representation for the decimal number -0.160123 . (06 Marks)
 c. Convert the Q-15 signed number 0.100011110110010 to the decimal number. (04 Marks)

OR

- 10 a. With a neat diagram, explain the fixed point basic architecture of TMS 320C54X processor. (10 Marks)
 b. Explain the IEEE double precision floating point format used in DSP processor. (05 Marks)
 c. Describe fixed point representation of numbers used in DSP processor. (05 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024
Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define amplitude modulation. Derive the expression of AM in both time domain and frequency domain representation with necessary waveforms. (08 Marks)
- b. With a neat block diagram, explain demodulation of DSB-SC wave using costas receiver. (06 Marks)
- c. With a neat block diagrams of a transmitter and receiver explain the operation of FDM system. (06 Marks)

OR

- 2 a. With a neat circuit diagram, waveforms and necessary equations, explain how the ring modulator can be used to generate DSB-SC wave. (10 Marks)
- b. An audio frequency signal $m(t) = 5\sin [2\pi \times 500t]$ is used to amplitude modulate the carrier of signal $c(t) = 10 \sin [2\pi \times 10^6t]$ find i) The modulation index μ ii) Side-band frequencies iii) Amplitude of each side band iv) Band width required v) Total power delivered to a load of 100Ω . (05 Marks)
- c. With a neat block diagrams of transmitter and receiver explain quadrature carrier multiplexing system. (05 Marks)

Module-2

- 3 a. Define and describe the time domain representation of frequency modulation and phase modulation with waveforms. (06 Marks)
- b. Explain the generation of narrow band FM with phasor diagram. (06 Marks)
- c. With relevant block diagrams, explain FM stereo multiplexing and demultiplexing. (08 Marks)

OR

- 4 a. Derive an expression for single-tone sinusoidal FM wave. (05 Marks)
- b. Explain with relevant block diagram, and mathematical expression, the demodulation of FM signal using non linear and linear model of the PLL. (10 Marks)
- c. The equation for a FM wave is $s(t) = 10\sin [5.7 \times 10^8t + 5 \sin 12 \times 10^3t]$. Find:
 - i) Carrier frequency
 - ii) Modulating frequency
 - iii) Modulation index
 - iv) Frequency deviation
 - v) Power dissipated in 100Ω . (05 Marks)

Module-3

- 5 a. With a neat block diagram of receiver model show that the figure of merit of for DSB-SC system is unity. (10 Marks)
- b. Discuss capture and threshold effect in FM receiving systems. (05 Marks)
- c. Show that the figure of merit of noisy AM receiver for single tone modulation is $\frac{\mu^2}{1 + \mu^2}$. (05 Marks)

OR

- 6 a. With a neat block diagram of receiver model using discriminator derive the expression for figure of merit of an FM receiver. (10 Marks)
- b. Explain the working of pre-emphasis and de-emphasis in frequency modulation. (06 Marks)
- c. Explain briefly the following: i) Shot noise ii) Thermal noise. (04 Marks)

Module-4

- 7 a. Mention the advantages of digital communication system. (04 Marks)
- b. Show that the signal $g(t)$ can be reconstructed using the interpolation formula shown below.

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \text{Sinc}(2Wt - n).$$
 (10 Marks)
- c. With a neat diagram, explain the concept of time division multiplexing. (06 Marks)

OR

- 8 a. Define the pulse amplitude modulation with relevant equations, waveforms, diagram, explain the generation and reconstruction of flat top sampling. (10 Marks)
- b. With a neat block diagram and wave forms, explain the generation of pulse position modulation waves. (10 Marks)

Module-5

- 9 a. With a neat block diagram, explain the concept of the PCM. (10 Marks)
- b. What is quantization process? Derive the expression for signal to quantization noise ratio for PCM signal where the input is sinusoidal signal. (10 Marks)

OR

- 10 a. With a neat encoder and decoder block diagram, necessary equations explain the delta modulation technique. (08 Marks)
- b. Represent the binary data: 10011101
 i) Polar NRZ ii) Bipolar RZ iii) Unipolar NRZ iv) Split phase formatting. (04 Marks)
- c. Write a short note on:
 i) Slope overload distortion
 ii) Granular noise. (08 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024

Electromagnetic Waves

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State and derive Coulomb's law. (06 Marks)
- b. Find the force on charge Q_1 located at (4, -2, 1) m due to charge Q_2 located at (3, -1, -2) m. $Q_1 = 200 \mu\text{C}$, $Q_2 = 300 \mu\text{C}$. (07 Marks)
- c. Calculate the electric field intensity E at (-1, 3, -2) m due to infinite line charges with $\rho_1 = 25 \text{ nC/m}$ lying along x-axis and $\rho_1 = 50 \text{ nC/m}$ lying along y-axis. (07 Marks)

OR

- 2 a. Derive electric field intensity E due to infinite line charge. (06 Marks)
- b. Two point charges $Q_1 = 5 \mu\text{C}$ and $Q_2 = -3 \mu\text{C}$ are located in free space at (1, 0, -2) m and (-2, 1, 3) m respectively. Find electric field intensity E at P(-3, 2, -1) m. (07 Marks)
- c. Calculate the electric field intensity E at (-2, 1, -3) m due to infinite sheet charges :
 $\rho_s = \frac{1}{6\pi} \text{ nC/m}^2$ located at $y = 3$ m and
 $\rho_s = \frac{1}{3\pi} \text{ nC/m}^2$ located at $z = -5$ m (07 Marks)

Module-2

- 3 a. State and prove Gauss law. (06 Marks)
- b. Given $D = \frac{5r^3}{y} \mathbf{a}_r \text{ C/m}^2$ in cylindrical co-ordinates. Prove divergence theorem for the volume enclosed by $r = 2\text{m}$, $r = 3\text{m}$, $z = 0$ and $z = 5\text{m}$. (07 Marks)
- c. Find the total charge in a volume defined by six planes for which, $2 \leq x \leq 3$, $3 \leq y \leq 4$, $4 \leq z \leq 5$, if $D = 5x^2 \mathbf{a}_x + 4y^2 \mathbf{a}_y + 3za_z \text{ C/m}^2$. (07 Marks)

OR

- 4 a. Using Gauss's law, derive the expressions for $\bar{D}$ and $\bar{E}$ due to co-axial cylindrical conductors. (06 Marks)
- b. Calculate the total electric flux density due to two uniform line charges of $30 \mu\text{C/m}$ lying along x-axis and $50 \mu\text{C/m}$ lying along z-axis, at (2, 3, 4)m. (07 Marks)
- c. In an electric field, potential field is $V = 5x^2 + 3y^3 + 8z$ volts. Find
 (i) $\bar{E}$ (ii) $|E|$ (iii) $\bar{D}$ at (-3, 2, 4) m (07 Marks)

Module-3

- 5 a. Using Laplace's equation, derive the expression for potential (V) and electric field strength E due to two concentric cylinders of infinite length. (06 Marks)
- b. In spherical co-ordinates $V = 750$ volts at $r = 25$ cm and $E = 825 \mathbf{a}_r \text{ V/m}$ at $r = 75$ cm. Determine the location of voltage reference if potential depends only on r . (07 Marks)
- c. State and prove Ampere's circuital law. (07 Marks)

OR

- 6 a. Using Biot-Savart's law, derive the expression for magnetic field intensity "H" due to infinite long conductor. (06 Marks)
- b. In spherical co-ordinates, $V = 0$ for $r = 0.2$ m and $V = 200$ volts for $r = 3$ m. Assuming free space between concentric spheres (Shells) find electric field intensity E and flux density D. (07 Marks)
- c. Find magnetic field intensity H at the center of a square loop of sides equal to 10 m and carrying a current of 5 amp. (07 Marks)

Module-4

- 7 a. Derive the equation for magnetic force on a differential current element in a magnetic field. (06 Marks)
- b. Calculate the force on a straight conductor of length 0.5 m carrying a current of 10 amp in the z-direction, where $\vec{B} = 5 \times 10^{-3} \hat{a}_x$ Tesla and $B = 6 \times 10^{-3} \hat{a}_y$ Tesla. (07 Marks)
- c. A solenoid with air core has 2000 turns and a length of 700 mm. Core radius is 50 mm. Find self inductance. (07 Marks)

OR

- 8 a. Derive the equation for force between two parallel current carrying conductors. (06 Marks)
- b. Derive tangential and normal boundary conditions (magnetic) between two media of permeabilities μ_1 and μ_2 . (07 Marks)
- c. Find the inductance per unit length of a co-axial conductor with an inner radius of $a = 4$ mm and outer radius of $b = 10$ mm. Assume $\mu_r = 1$. (07 Marks)

Module-5

- 9 a. State the inconsistency of Ampere's law, for time varying fields. Derive Maxwell's equation to correct it. (06 Marks)
- b. Derive general plane wave equation in terms of E, taking help of the Maxwell's equation (for free space). (07 Marks)
- c. A plane wave travelling in positive z-direction in a lossless unbounded medium has permeability 5 times that of free space and a dielectric constant 3 times that of free space.
- (i) Find phase velocity of the wave
- (ii) If E has only x-component with amplitude 25 V/m, find amplitude and direction of H. (07 Marks)

OR

- 10 a. Prove that conduction current and displacement current are equal. (06 Marks)
- b. State and explain Poynting theorem. (05 Marks)
- c. Determine following parameters for a medium with $\epsilon_r = 4$, $\mu_r = 1$, $\sigma = 20 \times 10^{-2}$ S/m, $f = 1$ mHz.
- Attenuation constant
 - Phase shift constant
 - Propagation constant
 - Wavelength
 - Phase velocity
 - Intrinsic impedance
 - Skin depth (δ)
- (09 Marks)

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18EC731

Seventh Semester B.E. Degree Examination, June/July 2024 Real Time Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Real Time System. Explain the classification of Real-Time System. (07 Marks)
- b. Explain the classification of programs. (08 Marks)
- c. Explain Hard and Soft Real Time System with an example. (05 Marks)

OR

- 2 a. Explain batch, continuous and laboratory system with an example. (09 Marks)
- b. Draw and explain direct digital control and its advantages. (08 Marks)
- c. Discuss PID algorithm. (03 Marks)

Module-2

- 3 a. Draw and explain single chip computer. (05 Marks)
- b. Explain the following: (i) SIMD (ii) MISC (iii) MIMD (09 Marks)
- c. Explain process related interfaces. (06 Marks)

OR

- 4 a. With a neat diagram, explain pulse input and output interfaces. (08 Marks)
- b. Explain digital input and output interface with diagram. (08 Marks)
- c. With flowchart, explain conditional transfer. (04 Marks)

Module-3

- 5 a. Explain the following with respect to real-time programming languages:
 - (i) Data type
 - (ii) Exception handling
 - (iii) Low-level facilities
 - (iv) Global and local variables(16 Marks)
- b. Explain the Co-Routines. (04 Marks)

OR

- 6 a. List and explain in brief the major requirements for real-time languages. (12 Marks)
- b. Explain the declaration and initialization of variables and constant in real-time languages. (08 Marks)

Module-4

- 7 a. With a neat diagram, explain multi-user operating system and multi-tasking operating system. (10 Marks)
- b. Explain the following:
 - (i) Priority levels
 - (ii) Scheduling strategies(10 Marks)

OR

- 8 a. Explain the RTOS state diagram. (10 Marks)
b. What is code sharing? Discuss the two methods of code sharing in detail. (10 Marks)

Module-5

- 9 a. Explain different phases involved in the design of RTS with diagram. (12 Marks)
b. With flow chart, explain the single program approach. (08 Marks)

OR

- 10 a. Explain Yourdon methodology. (05 Marks)
b. With a neat diagram, explain the outline of abstract modelling approach of Ward and Mellor. (07 Marks)
c. Explain the structure of the requirements model in Hatley and Pirabhai method. (08 Marks)

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18EC81

Eighth Semester B.E. Degree Examination, June/July 2024 Wireless and Cellular Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain path loss model for free space propagation. (06 Marks)
b. Explain briefly three basic propagation mechanisms. (06 Marks)
c. If a transmitter produces 50 W of power, express the transmit power in units of (i) dBm (ii) dBW. If 50 W is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna what is $P_r(10 \text{ km})$? Assume unity gain for the receiver antenna? (08 Marks)

OR

- 2 a. Distinguish between delay spread and coherence bandwidth. (06 Marks)
b. Distinguish between Doppler spread and coherence time. (06 Marks)
c. Explain the analysis of cellular systems. (08 Marks)

Module-2

- 3 a. Describe GSM protocols and signaling model with a neat diagram. (10 Marks)
b. Explain the various logical channels used in GSM. (10 Marks)

OR

- 4 a. List out the ten operations in call set up in GSM system. Explain in detail authentication and ciphering mode operation. (10 Marks)
b. Explain the intra BSC hand over operation in GSM. (10 Marks)

Module-3

- 5 a. Explain frequency planning issues for intersystems in CDMA. (08 Marks)
b. Explain the network nodes found in CDMA 2000 wireless system. (12 Marks)

OR

- 6 a. Explain basic spectrum spreading operation in CDMA. (10 Marks)
b. Explain the generation of the pilot channel signal. (10 Marks)

Module-4

- 7 a. List the advantages of OFDM leading to its selection for LTE and explain. (10 Marks)
b. With a neat block diagram, explain LTE network architecture and describe briefly the new elements provided in it. (10 Marks)

OR

- 8 a. With the help of neat diagrams, explain how the timing and frequency synchronization is performed by the receiver to demodulate an OFDM signal. (12 Marks)
b. What is PAR problem? Explain the methods used for PAR reduction. (08 Marks)

Module-5

- 9 a. Explain basic design principles followed in LTE specifications. (10 Marks)
b. Explain downlink OFDMA radio resources. (10 Marks)

OR

- 10 a. Explain uplink SC-FDMA radio resources. (10 Marks)
b. Explain the layers of LTE radio interface protocol. (10 Marks)

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18EC821

Eighth Semester B.E. Degree Examination, June/July 2024 Network Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain all the principles of security with suitable diagrams. (07 Marks)
- b. Explain Java Sand Box with related diagram which shows detailed steps in the execution of the java program on internet. (07 Marks)
- c. Write short notes on virus and worms. (06 Marks)

OR

- 2 a. Explain passive and active attacks with suitable diagrams. (07 Marks)
- b. List out two types of specific attacks and explain in detail. (07 Marks)
- c. What is cookie? Explain its creation and usage of cookies with relevant diagrams. (06 Marks)

Module-2

- 3 a. List the various web traffic security approaches and explain with relevant diagrams. (06 Marks)
- b. With suitable diagrams, explain the working of handshake protocol action. (08 Marks)
- c. With suitable diagram, explain SSL architecture with the concepts of connection and session. (06 Marks)

OR

- 4 a. What is transport layer security? Explain calculation of Message Authentication Code (MAC) and Generation of Pseudorandom function with suitable diagram. (10 Marks)
- b. Explain the working of HTTPS with related connection initiation and connection closure. (05 Marks)
- c. With suitable diagram, explain the packet formation of SSH transport layer protocol. (05 Marks)

Module-3

- 5 a. Explain IP Security overview with suitable diagram and list its applications. (07 Marks)
- b. Explain IPSec documents with suitable diagram and write a short note on security associations. (08 Marks)
- c. Explain how authentication header guards against the replay attack. (05 Marks)

OR

- 6 a. Explain the two ways in which IPSec authentication service can be used with related diagrams. (08 Marks)
- b. Explain the various fields of ESP format with suitable diagrams. (05 Marks)
- c. Explain the scope of ESP encryption and authentication in transport and tunnel mode with suitable frame format. (07 Marks)

Module-4

- 7 a. List and explain three classes of intruders. Explain various intrusion techniques. (10 Marks)
b. Define intrusion detection with suitable approaches. Explain statistical anomaly detection. (10 Marks)

OR

- 8 a. i) Define virus. Explain its life phases. (10 Marks)
ii) Explain virus structure with suitable example. (10 Marks)
b. Write short notes on:
(i) Digital immune system (10 Marks)
(ii) Antivirus approaches (10 Marks)

Module-5

- 9 a. List various types of firewalls. Explain the packet filtering router in detail. (10 Marks)
b. Explain various design goals of a fire wall. Also give details about the capabilities and limitations of firewall. (10 Marks)

OR

- 10 a. Define firewall configuration. Explain in detail the various configurations with suitable diagrams. (10 Marks)
b. Explain in detail the circuit level gateway with suitable diagrams. (10 Marks)

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21ELE13/23

First/Second Semester B.E. Degree Examination, June/July 2024 Basic Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State Ohms Law and mention its limitations. (05 Marks)
- b. Find the currents in the various branches of the given network shown in Fig.Q1(b).

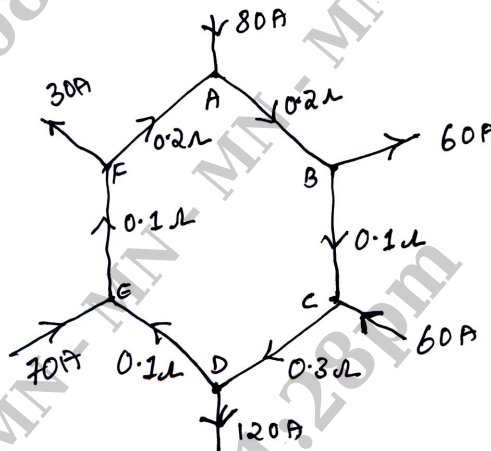


Fig.Q1(b)

- c. State and derive the condition to draw maximum power from the load using maximum power transfer theorem. (07 Marks)

OR

- 2 a. Define RMS value and average value of an alternating quantity and derive the expression for the same. (10 Marks)
- b. List the advantages of sinusoidal waveform. (05 Marks)
- c. Show that, the power consumed in a pure capacitance is zero. (05 Marks)

Module-2

- 3 a. Derive an expression for the power consumed in a series RL – AC circuit and draw voltage current and power waveform. (08 Marks)
- b. A circuit shown in Fig.Q3(b) consists of a resistance of 10Ω , an inductance of 16mH and a capacitance of $150\mu\text{F}$ connected in series. A supply of 100V , 50Hz is given to the circuit. Find the current, p.f and power consumed by the circuit.

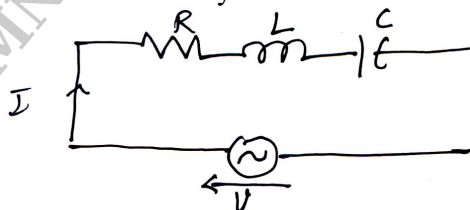


Fig.Q3(b)

- c. Draw the power triangle and define active power, reactive power and apparent power. (06 Marks)

(06 Marks)

OR

- 4 a. Derive the relationship between line and phase values of current in a three phase balanced star connected system. (06 Marks)
- b. A three phase load of three equal impedances connected in delta across a balanced 400V supply takes a line current of 10A, a power factor of 0.7 lagging. Calculate :
 i) The phase current
 ii) The total power
 iii) Total reactive KVA. (06 Marks)
- c. Show that two wattmeters are sufficient to measure 3 phase power and power factor of the circuit in a 3 phase balanced circuit. (08 Marks)

Module-3

- 5 a. Explain the construction of a DC generator with a neat sketch. (08 Marks)
- b. Derive an equation for the torque developed in the armature of a DC motor. (06 Marks)
- c. A 4-pole, 1500rpm DC generator has a lap wound armature having 24 slots with 10 conductors per slot. If the flux per pole is 0.04wb, calculate the emf generated in the armature. What would be the generated emf. If the winding is wave connected? (06 Marks)

OR

- 6 a. Derive the EMF equation of a transformer. (06 Marks)
- b. Explain different losses occurring in a transformer. (06 Marks)
- c. A transformer is rated at 100KVA, at full load its copper loss is 1200W and the iron loss is 960W calculate:
 i) The efficiency of full load, u.p.f.
 ii) The efficiency of at half load 0.8pf
 iii) The load KVA at which maximum efficiency occurs
 iv) Maximum efficiency at 0.85 pf. (08 Marks)

Module-4

- 7 a. Explain the concept of rotating magnetic field in case of a 3-phase induction motor. (08 Marks)
- b. Distinguish between wound rotor (slip ring) and squirrel cage rotor of induction motor. (06 Marks)
- c. Define slip of an induction motor and derive the relation between the supply frequency and rotor current frequency. (06 Marks)

OR

- 8 a. Derive the emf equation of alternator (Three phase synchronous generator). (08 Marks)
- b. Explain the construction of two types of alternator with a neat diagram. (06 Marks)
- c. A 3-phase 50Hz, 16-pole alternator with star connected winding has 144 slots with 10 conductors/slot. The flux/pole is 24.8mwb and the coils are full pitched. Find :
 i) The speed ii) Line EMF. Assume the distribution factor $k_d = 0.96$. (06 Marks)

Module-5

- 9 a. Explain the concept of power transmission and power distribution with a neat diagram of power system. (08 Marks)
- b. Explain tariff and list out the types of tariffs. (06 Marks)
- c. What are the desirable characteristics of a tariff and explain two-part tariff. (06 Marks)

OR

- 10 a. Explain the necessity of earthing and also explain pipe earthing with a neat diagram. (08 Marks)
- b. Explain the working RCCB with a neat diagram. (06 Marks)
- c. Write about precautions against electric shock. (06 Marks)

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First/Second Semester B.E. Degree Examination, June/July 2024 Basic Electronics and Communication Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe various electronic units of a d.c. power supply with a neat block diagram. (07 Marks)
- b. Derive an expression for overall gain of an amplifier with negative feedback. (07 Marks)
- c. Illustrate single-stage astable multivibrator using operational amplifier and explain the threshold voltages. (06 Marks)

OR

- 2 a. With neat block diagram and waveforms, explain the working of bridge rectifier. (08 Marks)
- b. Explain the working of op-amp comparators and summing amplifiers with input and output waveforms. (08 Marks)
- c. A 5 V zener diode has a maximum rated power dissipation of 500 mW. If the diode is to be used in a simple regulator circuit to supply a regulated 5 V to a load having resistance of $400\ \Omega$; determine a suitable value of series resistor for operation in conjunction with a supply of 9 V. (04 Marks)

Module-2

- 3 a. With the help of truth table and logic expressions, explain full adder using basic gates. (08 Marks)
- b. Discuss the design of a 3-bit asynchronous up-counter. (07 Marks)
- c. Write a note on different data types mentioning the bit size and range of values supported. (05 Marks)

OR

- 4 a. Design a 3-to-8 Decoder and show its implementation using basic gates. (07 Marks)
- b. With the help of timing diagram, explain how D-type bistable circuit works. (06 Marks)
- c. With a neat block diagram, show how typical input and output blocks are connected to a microcontroller unit. (07 Marks)

Module-3

- 5 a. Differentiate : Embedded systems versus General computing systems. Also provide major application areas of Embedded Systems. (08 Marks)
- b. Discuss arrangement of an instrumentation system and a control system. (06 Marks)
- c. Illustrate topology for USB device connection. Also, classify four different data transfers supported by USB. (06 Marks)

OR

- 6 a. Explain the principle of operation, working and applications of stepper motor. (08 Marks)
- b. With relevant diagrams, explain the operation of Relay. (06 Marks)
- c. Write a note on classification of Embedded systems. (06 Marks)

Module-4

- 7 a. Draw a block schematic diagram of the most general form of basic communication system and explain. (08 Marks)
- b. Explain different types of radio wave propagation with a neat diagram. (06 Marks)
- c. Differentiate : (i) Amplitude modulation versus Frequency modulation.
(ii) Analog modulation versus Digital modulation. (06 Marks)

OR

- 8 a. Explain PAM, PWM, PPM and PCM with the help of waveforms. (08 Marks)
- b. Discuss Forward Error Correction (FEC) technique. With neat diagram and example. (07 Marks)
- c. Define an antenna and discuss various types of antennas. (05 Marks)

Module-5

- 9 a. Draw the schematic diagram of cellular telephone system and define its basic components. (06 Marks)
- b. Draw the block diagram, showing the basic elements of a satellite communication system and briefly explain. (08 Marks)
- c. With the help of block diagram, explain generalized configuration of a fiber-optic communication system. (06 Marks)

OR

- 10 a. With a neat block diagram, explain GSM system architecture. (08 Marks)
- b. With the help of architecture figure, explain the evolution from GSM to LTE. (08 Marks)
- c. What is Bluetooth? Explain Bluetooth architecture. (04 Marks)

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21EC32

Third Semester B.E. Degree Examination, June/July 2024 Digital System Design using Verilog

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. The input to a combinational logic circuit is a valid single digit BCD data. Design the logic circuit using minimum hardware to detect whenever a number greater than 5 appears at the input. (08 Marks)
- b. Expand $f_1 = a + ab + a\bar{c}d$ into minterm and $f_2 = a.(b + c).(a + c + \bar{d})$ into maxterm. (06 Marks)
- c. Reduce the following function using K-map technique and implement using gates.
 $f(P, Q, R, S) = \sum m(0, 1, 4, 8, 9, 10) + \sum d(2, 11)$ (06 Marks)

OR

- 2 a. Identify all prime implicants and essential prime implicants of the following function using K-map.
 $f(a, b, c, d) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 13, 14, 15)$ (06 Marks)
- b. Simplify the following Boolean function using Quine McCluskey method.
 $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$ (10 Marks)
- c. Minimize the expression using K-map
 $Y = (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + B + C) \cdot (A + B + C)$ (04 Marks)

Module-2

- 3 a. Design 2-bit magnitude comparator. (10 Marks)
- b. Implement $f(w, x, y, z) = \sum m(0, 4, 8, 10, 14, 15)$ using
 - i) 8×1 MUX with w, x, y as select lines.
 - ii) 4×1 MUX with w, x as select lines.(06 Marks)
- c. Implement full adder using 74138 decoder. (04 Marks)

OR

- 4 a. Explain the general structure of PLDs. (06 Marks)
- b. Construct 4 to 16 line decoder from 2 to 4 line decoder and implement the Boolean function.
 $f(x_3, x_2, x_1, x_0) = \sum m(0, 6, 9, 11, 15)$ (08 Marks)
- c. Design 3 bit binary full subtractor using logic gates. (06 Marks)

Module-3

- 5 a. With neat diagram explain Master Slave JK Flip Flop. (08 Marks)
- b. Explain 4 bit universal shift register. (08 Marks)
- c. Obtain the characteristics equation for SR and T Flip Flop. (04 Marks)

OR

- 6 a. Explain the working of Mod-4 twisted ring counter. (07 Marks)
- b. Design 4 bit binary ripple counter using T Flip Flop. (06 Marks)
- c. Design Mod-6 synchronous counter using clocked JK Flip Flop. (07 Marks)

Module-4

- 7 a. With general syntax and suitable examples, explain the shift operators available in verilog. (06 Marks)
 b. List and explain the verilog data types. (08 Marks)
 c. Realize full adder circuit using verilog data flow description. (06 Marks)

OR

- 8 a. Write a verilog code for a 2×1 multiplexer in dataflow description using signal assignment. (06 Marks)
 b. List all the different styles of descriptions, explain the structure of dataflow description. (06 Marks)
 c. Explain the following in data flow description:
 i) Signal declaration and assignment statement
 ii) Constant declaration and assignment statement
 iii) Concurrent signal assignment statement
 iv) Assigning a delay time to the signal assignment statement. (08 Marks)

Module-5

- 9 a. Explain CASE statement with syntax. Write a behavioral description of a positive edge triggered JK Flip Flop using CASE statement in verilog. (10 Marks)
 b. Write verilog behavioral description of 8×1 MUX. (06 Marks)
 c. Write a verilog program for half adder using structural description. (04 Marks)

OR

- 10 a. With example, explain the syntax of following sequential statements:
 i) If ii) Else if (06 Marks)
 b. List and explain all the loop statements in verilog. (08 Marks)
 c. Write a verilog program for 3 bit binary counter using case statement. (06 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Basic Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Show that : W is a subspace of $V(F)$ iff.
- i) W is none empty
 - ii) $\forall a, b, \in F$ and $V_1 W \in W, [av + b.w] \in W.$ (06 Marks)
- b. Determine whether or not each of the following form a basis $x_1 = (2, 2, 1)$; $x_2(1, 3, 1)$, $x_3 = (1, 2, 2)$ in R^3 . (06 Marks)
- c. Evaluate u, v, w, are pair wise orthogonal vectors and find orthonormal vectors of
- $$u = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, w = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}.$$
- (08 Marks)

OR

- 2 a. Define vector subspace and explain the four fundamental subspace. (06 Marks)
- b. Determine the linear transformation of 'T' from $R^2 \rightarrow R^2$ such that $T(1, 0) = (1, 1)$ and $T(0, 1) = (-1, 2)$. (06 Marks)
- c. Apply Gram – Schemidt process to vectors, $V_1 = (1, 1, 1)$, $V_2 = (1, -1, 2)$, $V_3 = (2, 1, 2)$ to obtain an orthonormal basis for V_3 with the standard inner product. (08 Marks)

Module-2

- 3 a. Evaluate Eigen values and eigen vector for matrix :
- $$A = \begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}.$$
- (10 Marks)
- b. Factorize the matrix A into $A = U\Lambda V^T$ using single value decomposition :
- $$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}.$$
- (10 Marks)

OR

- 4 a. Diagonalize the matrix :
- $$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$
- (10 Marks)
- Find an invertible matrix P and diagonal matrix D such that $D = PAP^{-1}$. (06 Marks)
- b. Define positive definite matrix mention the methods of testing positive definite. (04 Marks)
- c. Determine eigen values and eigen vectors for
- $$A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}.$$

Module-3

- 5 a. Find and sketch : i) $y_1(n) = x(4 - n)$ ii) $y_2(n) = x(2n - 3)$ for given $x(n) = [u(n) - u(n - 8)]$. (07 Marks)
- b. Obtain whether the given system is linear, time invariance, memory causal :
i) $y_1(n) = n^2 x(n - 1)$, ii) $y_2(n) = \log_{10}[x(n)]$. (08 Marks)
- c. Describe the elementary signals. (05 Marks)

OR

- 6 a. Sketch : i) $Z(n) = x(2n)y(n - 4)$, given the signals $x(n)$ and $y(n)$ in the Fig.Q6(a).

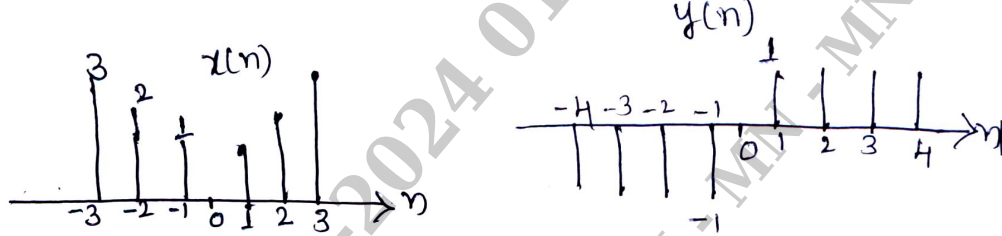


Fig.Q6(a)

- b. Check whether the following system is linear time invariance, memory causal.
i) $y(n) = x(n) + 2x^2(n)$, ii) $y(n) = g(n)x(n)$. (08 Marks)
- c. What is system? Explain its properties. (05 Marks)

Module-4

- 7 a. Evaluate $y(n) = x(n) * h(n)$. If $x(n) = \alpha^n u(n)$, $\alpha < 1$ and $h(n) = u(n)$. (06 Marks)
- b. Evaluate the step responses for the LTI system represented by the following impulse response : i) $h(n) = (1/2)^n u(n)$, ii) $h(n) = u(n)$. (06 Marks)
- c. Check whether given LTI system is stable, causal and compute the $h(n)$ for the sequence.
 $y(n) = x(n + 1) + 5x(n) - 7x(n - 1) + 4x(n - 2)$. (08 Marks)

OR

- 8 a. Evaluate the discrete time convolution sum $y(n) = (1/2)^n u(n - 2) * u(n)$. (10 Marks)
- b. Check whether the following system is memoryless, causal, stable
i) $h(n) = e^{2n} u(n - 1)$ ii) $h(n) = 2u(n) - 2u(n - 1)$. (10 Marks)

Module-5

- 9 a. State and prove the Differentiation in Z - domain in Z - Transformation. (05 Marks)
- b. Evaluate Z - Transform of given $x(n) = n \left(\frac{1}{2} \right)^n u(n)$. (08 Marks)
- c. Obtain Inverse Z - Transformation of given

$$X(Z) = \frac{-1 + 5Z^{-1}}{1 - \frac{3}{2}Z^{-1} + \frac{1}{2}Z^{-2}} \quad \text{ROC } |Z| > 1. \quad (07 \text{ Marks})$$

OR

- 10 a. Compute $H(z)$ and $h(n)$ for LTI system is described by
 $y(n) - \frac{1}{4}y(n-1) - \frac{3}{8}y(n-2) = -x(n) + 2x(n-1)$ (08 Marks)
- b. Obtain Z - Transformation of signal.
 $x(n) = 7 \left(\frac{1}{3} \right)^n u(n) - 6 \left(\frac{1}{2} \right)^n u(n)$. (07 Marks)
- c. What is ROC and list out the properties? (05 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024

Digital Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Prove that the sampling of Fourier transform of a sequence $x(n)$ results in N point DFT. Using which both the sequence and the transform can be reconstructed. (10 Marks)
- b. Compute the 8 point DFT of the sequence $x(n) = \{1, 1, 1, 1, 0, 0, 0, 0\}$. Also write a Matlab code to compute N point DFT of a sequence. (10 Marks)

OR

- 2 a. Compute the circular convolution of the given sequences using DFT and IDFT method
 $x_1(n) = \{2, 3, 1, 1\}$ and $x_2(n) = \{1, 3, 5, 3\}$. (10 Marks)
- b. Compute the N point DFT of $x(n) = \begin{cases} \frac{1}{3}; & 0 \leq n \leq 2 \\ 0; & \text{otherwise} \end{cases}$ (06 Marks)
- c. If $x(n) = \{1, 2, 0, 3, -2, 4, 7, 5\}$. Evaluate i) $X(0)$ ii) $X(4)$ iii) $\sum_{K=0}^7 X(K)$. (04 Marks)

Module-2

- 3 a. Determine the response of a LTI system with $h(n) = \{1, -1, 2\}$ for an input.
 $x(n) = \{1, 0, 1, -2, 1, 2, 3, -1, 0, 2\}$ using overlap add method. Use 6 point circular convolution in your approach. (10 Marks)
- b. Develop the 8 point DIF_FFT algorithm. Mention the property of phase factor exploited. (10 Marks)

OR

- 4 a. Determine 8 point DFT of $x(n) = \{1, 0, -1, 2, 1, 1, 0, 2\}$ using of radix-2 DIT-FFT algorithm. Clearly show all intermediate results. (10 Marks)
- b. State and prove circular time shift property. Also write the matlab code for the same. (10 Marks)

Module-3

- 5 a. Design a filter with

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega}; & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0; & \frac{\pi}{4} \leq |\omega| \leq \pi \end{cases}$$

Use Hamming window with $M = 7$. Obtain the system transfer function equation. (10 Marks)

- b. Consider a FIR filter with system function: $H(z) = 1 + 2.82z^{-1} + 3.4048z^{-2} + 1.74z^{-3}$. Sketch the direct form-I and lattice realization of the filter. (10 Marks)

OR

- 6 a. Write a Matlab code to design a high pass FIR filter, using hanning window. The expected output with necessary calculations to be shown. (10 Marks)
- b. Mention the two desirable characteristics of window function. Compare Rectangular, Hamming, Hanning and Bartlett window functions. (06 Marks)
- c. Given $H(z) = (1 + 0.6z^{-1})^3$. Realize as a cascade of 1st and 2nd order section. (04 Marks)

Module-4

- 7 a. Compare analog and digital filters. (04 Marks)
- b. Given $H(z) = \frac{8z^3 - 4z^2 + 11z - 2}{\left(z - \frac{1}{4}\right)\left(z^2 - z + \frac{1}{2}\right)}$. Realize in DF-I and DF-II. (06 Marks)
- c. Obtain the expression for order and cut-off frequency of Low Pass Butterworth filter. (10 Marks)

OR

- 8 a. Design a digital low pass filter using BLT method to satisfy the following characteristics:
 i) Monotonic stopband and pass band
 ii) -3db cut off frequency of 0.5π rad
 iii) Magnitude down atleast 15dB at 0.75π rad. (10 Marks)
- b. Mention two conditions of transforming the filter from s plane to z plane. Explain how is it achieved in bilinear transformation with mapping diagram. (06 Marks)
- c. Write a matlab code to design an analog LP Butterworth filter. (04 Marks)

Module-5

- 9 a. Explain :
 i) General Microprocessor based on Von Neumann architecture
 ii) Digital signal processors based on Harvard architecture. (12 Marks)
- b. Convert the following:
 i) Q15 signed number 0.100011110110010 to decimal number.
 ii) Decimal number -0.160123 to signed Q-15 representation. (08 Marks)

OR

- 10 a. Explain IEEE floating point formats. (10 Marks)
- b. Explain the basic architecture of TMS320C54X processor. (10 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024

Communication Theory

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive the expression in time domain and frequency domain for an AM wave. Outline the waveform and spectrum. (08 Marks)
 b. With neat diagram, explain costas receiver. (06 Marks)
 c. With relevant block diagram, explain the working of FDM transmitter and receiver. (06 Marks)

OR

- 2 a. With relevant equations and diagrams, explain the generation of AM waves using switching modulator. (08 Marks)
 b. Explain in detail quadrature carrier multiplexing and demultiplexing system. (06 Marks)
 c. An audio frequency signal $5 \sin 2\pi (1000)t$ is used to amplitude modulate a carrier of $100 \sin 2\pi (10^6)t$. Assume modulation index is 0.4. Find:
 i) Sideband frequencies
 ii) Amplitude of each sideband
 iii) Bandwidth required
 iv) Total power delivered to a load of 100Ω . (06 Marks)

Module-2

- 3 a. Define modulation index, frequency deviation and derive the time domain and frequency domain representation wideband FM with diagram. (08 Marks)
 b. With neat diagram, explain the FM demodulation using balanced slope detector/balanced frequency discriminator. (08 Marks)
 c. An FM signal has sinusoidal modulation with $W = 15\text{kHz}$ and modulation index $\beta = 2$. Using Carson's rule determine the transmission bandwidth and deviation ratio. Assume $\Delta f = 75\text{kHz}$. (04 Marks)

OR

- 4 a. With relevant diagram, explain direct method generation of FM using Hartley oscillator and how frequency stability is achieved. (08 Marks)
 b. With block diagram, explain the linear model of PLL. (06 Marks)
 c. With the aid of neat diagram, explain FM stereo multiplexing. (06 Marks)

Module-3

- 5 a. Explain shot noise and thermal noise with relevant diagrams and expressions. (06 Marks)
 b. Derive the equation for the figure of merit of an AM receiver and show figure of merit = 1/3 when operating on a single tone AM. (08 Marks)
 c. Explain about FM threshold effect and its reduction method. (06 Marks)

OR

- 6 a. With relevant diagrams and expressions explain noise equivalent bandwidth. (04 Marks)
 b. Derive the equation for the figure-of-merit for DSB-SC receiver is one. (08 Marks)
 c. Explain the pre-emphasis and de-emphasis with respect to FM system. (08 Marks)

Module-4

- 7 a. List the two operations involved in the generation of PAM and explain with neat waveform and derive the equations for flat-top sampled PAM. (10 Marks)
 b. Explain the working principle and operation of Time Division Multiplexing (TDM) with neat diagram. (10 Marks)

OR

- 8 a. State sampling theorem for a strictly band limited signal and explain how the effects of aliasing is overcome. (06 Marks)
 b. With neat diagram and equations explain the generation of PPM waves. (08 Marks)
 c. What are the advantages of transmission of digital information over analog information? (06 Marks)

Module-5

- 9 a. Discuss briefly quantization noise and show the output signal-to-noise ratio of a uniform quantizer is $(\text{SNR})_0 = \left[\frac{3P}{m_{\max}^2} \right] 2^{2R}$. (08 Marks)
 b. What is delta modulation? With neat block diagram, explain the construction of delta modulation. (06 Marks)
 c. Explain μ -law and A-law of compression. (06 Marks)

OR

- 10 a. Define quantization. Explain how the quantization process takes place. Illustrate the input output characteristics of a quantizer in terms of mid-tread and mid-rise. (08 Marks)
 b. With neat block diagram, explain the generation and reconstruction of PCM signals. (08 Marks)
 c. Write a short note on Vocoders. (04 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024 Digital Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat block diagram, explain the generation and detection of coherent QPSK signal. (10 Marks)
- b. Binary data is transmitted over a microwave link at a rate of 10^6 bits/sec and the PSD of noise at the receiver input is 10^{-10} watt/Hz. Find the average carrier power required to maintain an average probability of error $P_c \leq 10^{-4}$ for coherent binary FSK. What is the required channel bandwidth? Assume $\text{erf}(2.8) = 0.9998$. (10 Marks)

OR

- 2 a. With a neat block diagram, explain the generation and detection of non-coherent DPSK signal. (10 Marks)
- b. Derive an expression for error probability of binary PSK using coherent detection. (10 Marks)

Module-2

- 3 a. Explain error-Schmidt orthogonalization procedure. (10 Marks)
- b. Explain matched filter receiver. List the properties of matched filter receiver. (10 Marks)

OR

- 4 a. Three signals $S_1(f)$, $S_2(f)$ and $S_3(f)$ are shown in Fig. Q4 (a) below. Apply Gram-Schmidt procedure to obtain an orthonormal basis for signals. Express $S_1(f)$, $S_2(f)$ and $S_3(f)$ in terms of orthonormal basis functions. (10 Marks)

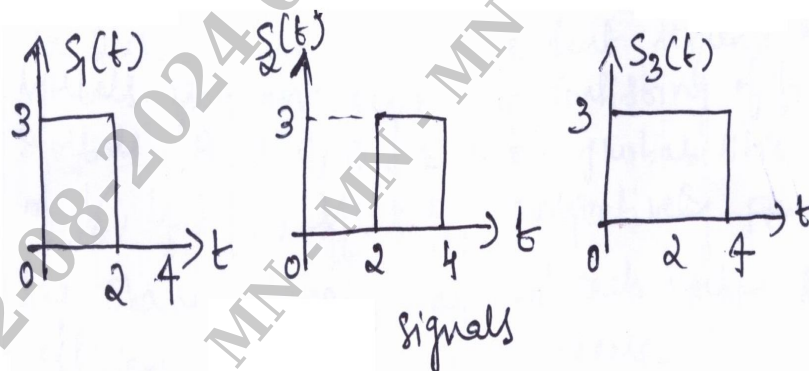


Fig. Q4 (a)

- b. Explain the design of bandlimited signals, with controlled ISI-partial response. (10 Marks)

Module-3

- 5 a. Explain the model of spread spectrum digital communication. (10 Marks)
- b. Explain direct sequence spread-spectrum system. (10 Marks)

OR

- 6 a. For a linear feedback shift register with three stage ($M = 3$), evaluate the maximum length PN sequence for feedback taps = (3, 1). Draw the schematic arrangement and verify all the properties of PN sequence is generated output. Sketch the sequence, its autocorrelation function. If the chip rate happens to be 10 MHz. Test all the three properties of ML sequence after generating PN sequence for a 3-stage feedback shift register. Assume 100 as initial state. (10 Marks)
- b. Explain the application of DS-spread spectrum signals. (10 Marks)

Module-4

- 7 a. State the properties of entropy. Derive an expression for average information content of symbols in long independent sequences. (10 Marks)
- b. A source emits an independent sequence of symbols from an alphabet consisting of five symbols. A, B, C, D and E with probabilities of $\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{3}{16}$ and $\frac{5}{16}$ respectively. Find the Shannon binary code for each symbol and efficiency of the coding scheme. (10 Marks)

OR

- 8 a. Consider the zero memory source with probabilities $P = \{0.4, 0.2, 0.1, 0.1, 0.05, 0.05\}$
- (i) Construct a binary Huffman code by placing the composite symbol as low as possible.
- (ii) Construct binary Huffman code by placing composite symbol as high as possible. In each case compute the variances of the word length and comment on the result. (10 Marks)
- b. Explain the types of codes. (10 Marks)

Module-5

- 9 a. Define Hamming weight, Hamming distance and Minimum distance. (06 Marks)
- b. Explain single error correcting Hamming codes. (06 Marks)
- c. What do you understand by Trelli's diagram? Explain clearly. (08 Marks)

OR

- 10 a. Explain Matrix description of Linear block codes. (10 Marks)

- b. For a systematic (7, 4) LBC, the parity matrix P is given by $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.
- (i) Find all possible valid code-vectors.
- (ii) Draw the corresponding encoding circuit.
- (iii) A single error has occurred within the given received vectors. Detect and correct those errors, $R_A = 0111110$ and $R_B = 1011100$
- (iv) Draw the syndrome calculation circuit. (10 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024

Research Methodology and Intellectual Property Rights

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Identify the meaning of Research and brief out the objectives and motivation in Engineering Research. (10 Marks)
- b. Identify the steps to find the "Solve a worthwhile problem". (10 Marks)

OR

- 2 a. Identify the types of Engineering Research and briefly explain them. (10 Marks)
- b. Identify the ethical issues related to authorship and brief out them. (10 Marks)

Module-2

- 3 a. Identify the essence of new and existing knowledge and explain briefly. (10 Marks)
- b. Identify how search operators are used to narrow down the search results. (10 Marks)

OR

- 4 a. Identify the impacts of title and keywords on citation. (10 Marks)
- b. Identify acknowledgement and attributions in research process and briefly explain. (10 Marks)

Module-3

- 5 a. Define Intellectual Property (IP). Explain the major types of IP. (10 Marks)
- b. Identify the process of patenting. Briefly explain. (10 Marks)

OR

- 6 a. Explain briefly the Commercialization of a patent. (10 Marks)
- b. What are the exclusions (product and processes) that cannot be patented? Explain. (10 Marks)

Module-4

- 7 a. Explain the classes or types of copyrights. (10 Marks)
- b. What is a Trademark? Explain the symbols in TM. (10 Marks)

OR

- 8 a. What are the advantages "the registration of a trademark provides to the proprietor"? (10 Marks)
- b. Identify process of Trademark registration and explain briefly the classification of TM. (10 Marks)

Module-5

- 9 a. Define Industrial design. Briefly explain acts and laws to govern Industrial design. (10 Marks)
- b. Identify procedure for registration of Industrial design by taking example of Aple Inc Vs Samsung Electronics Co. (10 Marks)

OR

- 10 a. Define Geographical Identification (GI) and briefly explain acts, laws and rules pertaining to GI. (10 Marks)
- b. Identify IP Organizations in INDIA. Explain schemes and programs for Intellectual Property Rights. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

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Sixth Semester B.E. Degree Examination, June/July 2024

Technological Innovation Management & Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Management. Explain the various functions of management. (10 Marks)
- b. Discuss and justify management as a Science, Art and Profession. (10 Marks)

OR

- 2 a. Explain Types, Steps and Limitation of planning. (10 Marks)
- b. What are managerial skills? Explain. (06 Marks)
- c. Distinguish between Management and Administration. (04 Marks)

Module-2

- 3 a. Define Organization. Briefly explain the principle of organizing. (10 Marks)
- b. What is Departmentalization? Explain process and purpose of Departmentalization. (10 Marks)

OR

- 4 a. With the help of diagram, explain Maslows need hierarchy theory with examples. (10 Marks)
- b. Discuss the requirements of effective direction. (05 Marks)
- c. Explain the importance of communication. (05 Marks)

Module-3

- 5 a. Explain functions of a Leader. (10 Marks)
- b. What is the behavioural approach of Leadership? Explain. (10 Marks)

OR

- 6 a. Explain the different views of social responsibility of a business. (10 Marks)
- b. Write a short note on need for control system. (05 Marks)
- c. What is Social Audit? Explain its benefits and limitation. (05 Marks)

Module-4

- 7 a. What is Entrepreneurship? Explain the functions of an Entrepreneurship. (10 Marks)
- b. Explain the various types of family Business. (04 Marks)
- c. List the problem faced by an Entrepreneur. (06 Marks)

OR

- 8 a. How to Identify the business opportunity? Explain. (10 Marks)
- b. Describe the various methods used for Generating business ideas. (10 Marks)

Module-5

- 9 a. What is Business plan? Explain the purpose of business plan. (10 Marks)
b. Explain the National board for micro, small and medium Enterprises (NBMSME) with its objective and scope. (10 Marks)

OR

- 10 a. Write a short note on :
(i) State Industrial Development Corporation (SIDCs).
(ii) State industrial area development board (SIADB) (10 Marks)
b. Explain the functions of,
(i) Small Industries Development Bank of India (SIDBI).
(ii) Technical Consultancy Organization (TCO). (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Microwave Theory and Antennas

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of drift velocity graph and wave from, explain the constructional feature and working of n-type GaAs diode. (10 Marks)
- b. A transmission line has the following primary constants $R = 2 \Omega/m$, $L = 8 \text{ nH/m}$, $G = 0.5 \text{ mS/m}$, $C = 0.23 \text{ pF/m}$ and $f = 1 \text{ GHz}$. Find :
 - (i) Characteristic impedance Z_0 .
 - (ii) Propagation constant γ
 - (iii) Wavelength λ .
 - (iv) Phase velocity V_p (10 Marks)

OR

- 2 a. Derive the expression for the voltage of current at any point on the transmission line equation and solution starting from the fundamentals. (10 Marks)
- b. Explain the standing waves with neat waveforms. (10 Marks)

Module-2

- 3 a. Derive scattering parameters for a multiport network. (10 Marks)
- b. The transmission lines of characteristic impedances Z_1 and Z_2 are joined at plane PP' . Express S-parameters in terms of impedances. (10 Marks)

OR

- 4 a. Derive S-matrix for a Magic Tee with neat diagram and its applications. (10 Marks)
- b. Explain the working of precision Dielectric Rotary phase shifter. (10 Marks)

Module-3

- 5 a. Discuss the operation of micro strip lines with its structure. Compare strip line and microstrip line. (10 Marks)
- b. Explain the operation of parallel strip line along with a neat diagram. Write down the expression for characteristic impedance. (10 Marks)

OR

- 6 a. Explain the following terms as related to antenna system :
 - (i) Directivity and gain.
 - (ii) Beam area.
 - (iii) Effective height
 - (iv) Bandwidth (10 Marks)
- b. A radio link has a 15 W transmitter connected to an antenna of 2.5 m^2 effective aperture at 5 GHz. The receiving antenna has an effective aperture 0.5 m^2 and is located 15 km line of sight distance from the transmitting antenna. Assuming lossless, matched antenna, find the power delivered to the receiver. (10 Marks)

Module-4

- 7 a. Explain the field pattern and phase pattern with a neat diagram. (10 Marks)
 b. Derive an expression and draw the field pattern for an array of two isotropic point sources situated symmetrical with respect to origin with equal amplitude and phase spaced $\frac{\lambda}{2}$ apart. (10 Marks)

OR

- 8 a. Derive an expression for field of a dipole in general for the case of thin linear antenna. (10 Marks)
 b. Find the directivity D for the sources with radiation intensity :
 (i) $U = U_m \sin^2 \theta$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$
 (ii) $U = U_m \cos^2 \theta$, $0 \leq \theta \leq \frac{\pi}{2}$, $0 \leq \phi \leq 2\pi$ (10 Marks)

Module-5

- 9 a. Derive an expression for field strength E_ϕ and H_ϕ in case of small loop antenna. (10 Marks)
 b. Derive an expression for radiation resistance of a small loop antenna. (10 Marks)
- OR**
- 10 a. Derive an expression for radiation resistance of a short dipole antenna. (10 Marks)
 b. Explain the different types of horn antenna with a diagram. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 VLSI Design and Testing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State Moore's law, elaborate with graph. (04 Marks)
- b. Derive the expression for drain current in linear and saturation regions. (10 Marks)
- c. Explain the following non-ideal characteristics: (06 Marks)
 - i) Body effect ii) Channel length modulation.

OR

- 2 a. Explain working of nMOS enhancement mode transistor operation with neat sketches and relevant equations. (08 Marks)
- b. Draw inverter circuit and explain its DC transfer characteristics. (08 Marks)
- c. Draw the schematic of (04 Marks)
 - i) $F = A + BC$ ii) $F = \overline{AB + CD}$.

Module-2

- 3 a. Explain CMOS fabrication process with necessary diagrams. (10 Marks)
- b. With relevant equations explain transient response of CMOS inverter. (05 Marks)
- c. With neat diagrams explain layout design rules. (05 Marks)

OR

- 4 a. Draw the stick diagram and layout of three input NAND gate. (06 Marks)
- b. Find maximum and minimum rise time and fall time delays of two input NAND gate. (06 Marks)
- c. Estimate the minimum delay of the path from A to B in Fig.Q.4(c) and choose transistor sizes to achieve this delay. The initial NAND2 gate may present a load of 8λ of transistor width on the input and the output load is equivalent to 45λ of transistor width.

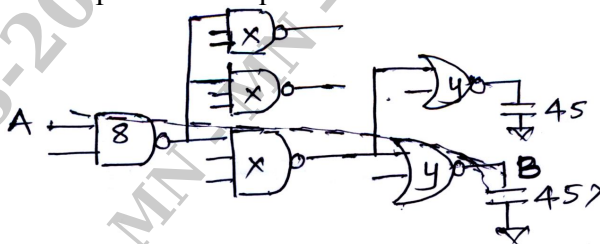


Fig.Q.4(c)

(08 Marks)

Module-3

- 5 a. Explain the operation of three transistor DRAM cell with necessary timing diagrams. (08 Marks)
- b. Draw the structure of NAND flash memory cell and explain the operation. (06 Marks)
- c. Explain ferroelectric RAM with necessary diagrams. (06 Marks)

OR

- 6 a. Explain read and write operations of SRAM cell with necessary diagrams. (08 Marks)
 b. What is row decoder? Explain with an example. (06 Marks)
 c. Explain data programming and erasing methods of flash memory. (06 Marks)

Module-4

- 7 a. Briefly explain different types of faults in digital circuits. (08 Marks)
 b. Consider the logic circuit shown in Fig.Q.7(b) find Boolean difference with respect to x_3 .

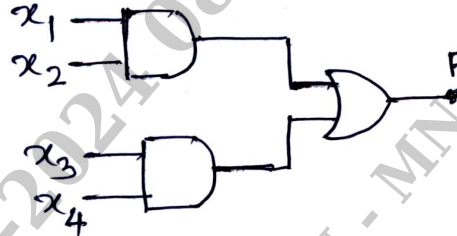


Fig.Q.7(b)

- c. Explain detection of multiple faults in combinational logic circuits. (06 Marks)

OR

- 8 a. With neat sketch, explain path oriented decision making algorithm. (10 Marks)
 b. For a given logic network determine tests for checking all single node faults (Fig.Q.8(b))

$$F = \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2 x_3$$

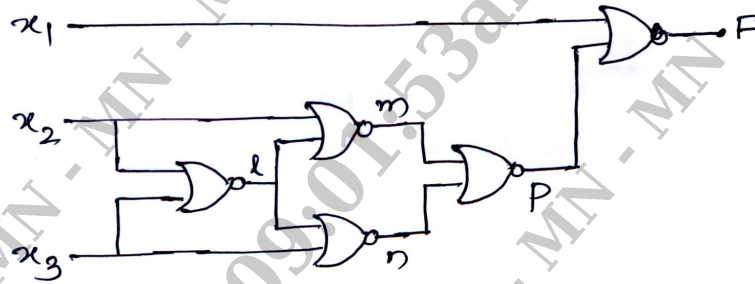


Fig.Q.8(b)

Module-5

- 9 a. Briefly explain :
 i) Controllability ii) Observability. (06 Marks)
 b. Explain adhoc design rules for improving testability. (06 Marks)
 c. With neat diagram explain partial scan. (08 Marks)

OR

- 10 a. List LSSD design rules. (10 Marks)
 b. Explain test generation based on functional fault models. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024

Cryptography

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the division algorithm with an example. (07 Marks)
- b. Define Ring. State six properties of Rings. (07 Marks)
- c. Explain the Euclidean algorithm. Calculate the GCD(60, -24). (06 Marks)

OR

- 2 a. Lists the properties of modular arithmetic for integer in 2^n with expression. (07 Marks)
- b. Explain the polynomial arithmetic. Find polynomial arithmetic over GF(2) for $f(x) = x^7 + x^5 + x^4 + x^3 + x + 1$ and $g(x) = x^3 + x + 1$. (07 Marks)
- c. Develop set of tables for polynomial arithmetic modulo of $x^3 + x + 1$ over GF(2³). (06 Marks)

Module-2

- 3 a. Draw and explain model of symmetric encryption. (07 Marks)
- b. Explain the playfair cipher and its rules for the following keyword : "MONARCHY" plaintext : "Cryptography". (07 Marks)
- c. Explain the vernam Cipher with a neat diagram. (06 Marks)

OR

- 4 a. Draw and explain model of symmetric cryptosystem. (07 Marks)
- b. Explain the Caesar Cipher technique Encrypt plaintext "Cryptography" with key = 3. (06 Marks)
- c. Using Hill Cipher algorithm Encrypt the plaintext "paymoremoney" using the key,

$$K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

(07 Marks)

Module-3

- 5 a. State and prove Euler's theorem. (05 Marks)
- b. Explain the DES encryption algorithm with neat diagram. (10 Marks)
- c. Explain Block Cipher with neat diagram. (05 Marks)

OR

- 6 a. Explain Feistel encryption and decryption with neat diagram. (10 Marks)
- b. State and prove Fermat's theorem. (05 Marks)
- c. Explain Euler's Totient function. Determine (i) $\phi(37)$ and $\phi(35)$. (05 Marks)

Module-4

- 7 a. Bring out differentiate between conventional encryption and public-key encryption. Explain the requirement of public-key cryptography. (10 Marks)
- b. Explain RSA algorithm. Using RSA algorithm perform encryption and decryption using $p = 17$, $q = 11$, $e = 7$ and $M = 88$. (10 Marks)

OR

- 8 a. Explain the Diffie-Hellman key exchange algorithm. Evaluate a Diffie-Hellman key exchange for $q = 23$ and $\alpha = 9$.
- (i) If User A has private key $X_A = 4$
What is A's public key $Y_A = ?$
 - (ii) If User B has private key $X_B = 3$
What is B's public key $Y_B = ?$
 - (iii) What is shared key ?
- b. Describe Elgamal cryptographic system.

(10 Marks)

(10 Marks)

Module-5

- 9 a. Write short notes on, (i) NANOTEQ (ii) A5 (iii) Linear Congruential generator.
- b. Explain Additive generator.
- c. With a neat diagram, explain Threshold generator.

(10 Marks)

(06 Marks)

(04 Marks)

OR

- 10 a. Explain linear feedback shift register with a neat diagram.
- b. With a neat diagram, explain Geffe generator and Jennings generator.
- c. Explain Gifford with a neat diagram.

(06 Marks)

(10 Marks)

(04 Marks)

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BBEE103/203

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024

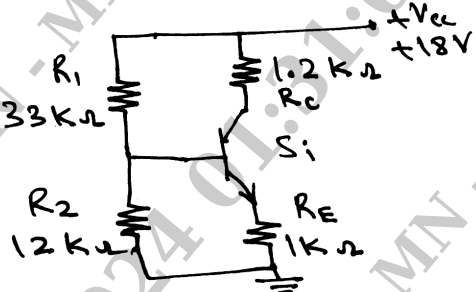
Basic Electronics for EEE Stream

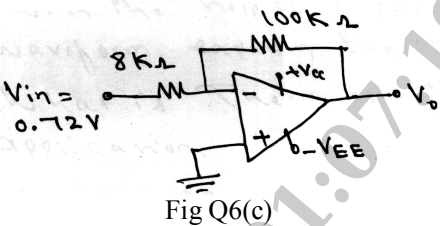
Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Sketch the forward and Reverse characteristics for a Silicon diode and explain it.	6	L2	CO1
	b.	Explain the working of a Half wave rectifier with input and output waveform.	8	L2	CO1
	c.	What is Filter? Mention the types of Filter.	6	L1	CO1
OR					
Q.2	a.	Write the various Diode Approximations.	8	L1	CO1
	b.	With circuit diagram and waveform, explain the working of a RC – π filter using Bridge rectifier.	6	L2	CO1
	c.	Explain the working of a Zener diode as a voltage Regulator with no load.	6	L2	CO1
Module – 2					
Q.3	a.	Calculate I_c , I_E , and β for a Transistor that has $\alpha = 0.98$ and $I_B = 100\mu A$.	6	L3	CO2
	b.	Draw the input and output characteristics of a common – Emitter of a Transistor and explain it.	8	L2	CO2
	c.	For the voltage divider bias circuit shown in Fig Q2(c), determine V_B , V_E , I_E and V_{CE} . Assume $V_{BE} = 0.7V$ 	6	L3	CO2
OR					
Q.4	a.	Explain the operation of an n-channel JFET for various bias voltages.	6	L2	CO2
	b.	Mention the advantages of FET over a BJT.	6	L1	CO2
	c.	Explain the construction of Enhancement MOSFET.	8	L2	CO2
Module – 3					
Q.5	a.	Mention the ideal characteristics of Op-Amp.	10	L2	CO2
	b.	Define the following parameters of Op-Amp i) CMRR ii) Slew rate iii) PSRR iv) Input offset voltage.	4	L1	CO2
	c.	Derive the expression of voltage Gain of a Non-inverting Op-Amp.	6	L2	CO2
OR					
Q.6	a.	How Op-Amp can be used as an integrator.	6	L2	CO2

	b.	Draw the block diagram of Typical Op-Amp and mention the function of each block.	6	L1	CO2
	c.	For the circuit shown in Fig Q6(c), find output voltage and voltage gain.  Fig Q6(c)	8	L3	CO2
Module – 4					
Q.7	a.	Perform the following : i) $(532.65)_{10} = ()_{16} = ()_2$ ii) $(ABCD)_{16} = ()_2 = ()_8$.	8	L3	CO3
	b.	State and prove the De Morgan's theorem for two variables.	8	L1	CO3
	c.	Using basic Boolean theorem prove that $(x + y)(x + z) = x + yz$.	4	L3	CO3
OR					
Q.8	a.	Draw the logic circuit for the Boolean expression $Y = \overline{A}BC + A\overline{B}C + ABC$.	4	L1	CO4
	b.	Implement full adder using Two half adder and an OR-Gate.	8	L3	CO4
	c.	Simplify the following Boolean expressions $(A + B)(\overline{A} + B)$ ii) $\overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}\overline{C}$.	8	L3	CO4
Module – 5					
Q.9	a.	What is strain Gauge? Explain the construction of unbounded strain gauge.	8	L2	CO5
	b.	With the help of circuit diagram and waveform, explain the operation of LVDT.	8	L2	CO5
	c.	What is Thermistor? Mention its applications.	4	L1	CO5
OR					
Q.10	a.	Explain the working of Photodiode.	7	L2	CO5
	b.	Draw the block diagram of Superhetrodyne receiver and mention the function of each block.	10	L2	CO5
	c.	Mention the need for modulation.	3	L1	CO5

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
AV Mathematics – III for EC/BM Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book and statistical table are permitted.

3. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1				M	L	C															
Q.1	a.	Find the Fourier series for $f(x)=\begin{cases} -K, & \text{in } (-\pi,0) \\ K & \text{in } (0,\pi) \end{cases}$ and hence deduce $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+.....$	6	L2	CO1																
	b.	Expand $f(x)=2x-1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																
	c.	Express y as a Fourier series upto the first harmonics given the following values: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>4</td><td>8</td><td>15</td><td>7</td><td>6</td><td>2</td></tr></table>	x	0	1	2	3	4	5	y	4	8	15	7	6	2	7	L3	CO1		
x	0	1	2	3	4	5															
y	4	8	15	7	6	2															
OR																					
Q.2	a.	Find the Fourier series for $f(x)=x-x^2$ in $-1 < x < 1$.	6	L2	CO1																
	b.	Show that half range sine series of $f(x)=\pi x-x^2$ in the interval $(0,\pi)$ is $\frac{8}{\pi}\sum_{h=0}^{\infty}\frac{1}{(2n+1)^3}\sin(2n+1)x$	7	L2	CO1																
	c.	Obtain the Fourier series of y upto 2 nd harmonics f(x) is given by <table><tr><td>x</td><td>0</td><td>$\pi/3$</td><td>$2\pi/3$</td><td>π</td><td>$4\pi/3$</td><td>$5\pi/3$</td><td>2π</td></tr><tr><td>f(x)</td><td>1.98</td><td>1.30</td><td>1.05</td><td>1.30</td><td>-0.88</td><td>-0.25</td><td>1.98</td></tr></table>	x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π	f(x)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98	7	L3	CO1
x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π														
f(x)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98														
Module – 2																					
Q.3	a.	Find the Fourier transform of $f(x)=\begin{cases} 1-x^2, & x <1 \\ 0 & x \geq1 \end{cases}$ and hence find the value of $\int_0^{\infty}\frac{x\cos x-\sin x}{x^3}dx$	6	L2	CO2																
	b.	Find the Fourier sine and cosine transform of $f(x)=e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																
	c.	Solve the integral equation $\int_0^{\infty}f(\theta)\cos\alpha\theta d\theta=\begin{cases} 1-\alpha, & 0\leq\alpha\leq1 \\ 0, & \alpha>1 \end{cases}$ and hence evaluate $\int_0^{\infty}\frac{\sin^2 t}{t^2}dt$.	7	L3	CO2																

1 of 3

OR

Q.4	a.	Find the Fourier transform of $e^{-a^2x^2}$, $a > 0$.	6	L2	CO2
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^\infty \frac{x \sin mx}{1+x^2} dx$, $m > 0$	7	L2	CO2
	c.	Find the discrete Fourier transform of the sequence $\{1, 2, 1, 3\}^T$.	7	L3	CO2

Module – 3

Q.5	a.	Obtain the Z-transform i) $\cos n\theta$ ii) $\sin n\theta$.	6	L2	CO3
	b.	Find the inverse Z-transform of $\frac{3z^2 + 2z}{(5z-1)(5z+2)}$	7	L2	CO3
	c.	Solve by using Z-transforms : $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$.	7	L3	CO3

OR

Q.6	a.	Find the Z-transform of $2n + \sin\left(\frac{n\pi}{4}\right) + 1$	6	L2	CO3
	b.	Find the inverse Z-transform of $\frac{4z^2 - 2z}{(z-1)(z-2)^2}$.	7	L2	CO3
	c.	If $\bar{u}(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$ find the value of u_0, u_1, u_2 .	7	L3	CO3

Module – 4

Q.7	a.	Solve $(D^4 + 8D^2 + 16)y = 0$.	6	L1	CO4
	b.	Solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 13y = e^{3t} \cosh 2t$.	7	L2	CO4
	c.	Solve $x^3 + x^2y'' + xy' + 8y = 65 \cos(\log x)$.	7	L3	CO4

OR

Q.8	a.	Solve $y'' + 9y = \cos 2x \cos x$.	6	L2	CO4
	b.	Solve $(2x+1)^2 y'' - 2(2x+1)y' - 12y = 6x + 5$.	7	L2	CO4
	c.	In an LCR circuit, the charge q on a plate of a condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{c} = E \sin pt$. Solve the equation for q .	7	L3	CO4

Module – 5

Q.9	a.	Fit a straight line for the following data: <table><tr><td>x</td><td>50</td><td>70</td><td>100</td><td>120</td></tr><tr><td>y</td><td>12</td><td>15</td><td>21</td><td>25</td></tr></table>	x	50	70	100	120	y	12	15	21	25	6	L1	CO5												
x	50	70	100	120																							
y	12	15	21	25																							
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>9</td><td>8</td><td>10</td><td>12</td><td>11</td><td>13</td><td>14</td></tr></table>	x	1	2	3	4	5	6	7	y	9	8	10	12	11	13	14	7	L2	CO5						
x	1	2	3	4	5	6	7																				
y	9	8	10	12	11	13	14																				
	c.	Compute the rank correlation coefficient for the following data: <table><tr><td>x</td><td>68</td><td>63</td><td>75</td><td>50</td><td>62</td><td>80</td><td>78</td><td>40</td><td>55</td><td>60</td></tr><tr><td>y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>	x	68	63	75	50	62	80	78	40	55	60	y	62	58	68	45	81	60	68	48	50	70	7	L3	CO5
x	68	63	75	50	62	80	78	40	55	60																	
y	62	58	68	45	81	60	68	48	50	70																	
OR																											
Q.10	a.	An experiment on life time 't' of cutting tool at different cutting speeds v(units) are given below <table><tr><td>Speed (v)</td><td>350</td><td>400</td><td>500</td><td>600</td></tr><tr><td>Life (t)</td><td>61</td><td>26</td><td>7</td><td>2.6</td></tr></table> Fit a relation of the form $v = at^b$.	Speed (v)	350	400	500	600	Life (t)	61	26	7	2.6	6	L2	CO5												
Speed (v)	350	400	500	600																							
Life (t)	61	26	7	2.6																							
	b.	The following data gives the age of husband (x) and the age of wife (y) in years. Form the 2 regression lines and calculate the age of husband corresponding to 16 years of age of wife. <table><tr><td>x</td><td>36</td><td>23</td><td>27</td><td>28</td><td>28</td><td>29</td><td>30</td><td>31</td><td>33</td><td>35</td></tr><tr><td>y</td><td>29</td><td>18</td><td>20</td><td>22</td><td>27</td><td>21</td><td>29</td><td>27</td><td>29</td><td>28</td></tr></table>	x	36	23	27	28	28	29	30	31	33	35	y	29	18	20	22	27	21	29	27	29	28	7	L2	CO5
x	36	23	27	28	28	29	30	31	33	35																	
y	29	18	20	22	27	21	29	27	29	28																	
	c.	If the coefficient of correlation between the variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}(3/5)$. Show that $\sigma_y = 2\sigma_x$.	7	L3	CO5																						

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Digital System Design using Verilog

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define combinational logic. Give two examples.	4	L1	CO1
	b.	Explain the procedure to place a sum of products equation into canonical form. Express the function $P = f(a, b, c) = ab' + ac' + bc$ in canonical form.	6	L2	CO1
	c.	Solve the function $K = f(w, x, y, z) = \Sigma(0, 1, 4, 5, 9, 11, 13, 15)$ using Karnaugh map.	4	L2	CO1
	d.	Simplify the function $F = f(P, Q, R, S) = \Sigma m(0, 3, 5, 6, 7, 11, 14)$ using Quine-McCluskey method.	6	L2	CO1
OR					
Q.2	a.	Define canonical sum of products and canonical product of sums. Give examples.	4	L1	CO1
	b.	Explain the procedure to place a product of sums equation into canonical form. Explain the function $T = f(a, b, c) = (a + b')(b' + c)$ in canonical form.	6	L2	CO1
	c.	Solve the function $G = f(a, b, c, d) = \pi(0, 4, 5, 7, 8, 9, 11, 12, 13, 15)$ using Karnaugh map.	4	L2	CO1
	d.	Simplify the function $F = f(P, Q, R, S) = \Sigma m(1, 2, 3, 5, 9, 10, 12)$ using Quine-McCluskey method.	6	L2	CO1
Module – 2					
Q.3	a.	Define encoder. Write the truth table, equations and circuit diagram of 8 – to – 3 – line priority encoder.	4	L1	CO2
	b.	Explain the concept of carry-lookahead adder with related equations and block diagram.	6	L2	CO2
	c.	Design one-bit comparator with inputs A_i, B_i – bits to be compared, G_i, E_i, L_i – previous stage inputs and with the outputs $G_{i+1}, E_{i+1}, L_{i+1}$.	6	L4	CO2
	d.	Implement the function $f(w, x, y, z) = \Sigma m(0, 1, 5, 6, 7, 9, 12, 15)$ using 8 – to – 1 – line multiplexer.	4	L3	CO2
1 of 3					

OR

Q.4	a.	Define decoder. Write the truth table, equations and circuit diagram of 3 – to – 8 – line decoder.	4	L1	CO2
	b.	Explain the operation of 8 – to – 1 line multiplexer with block diagram, truth table, equation.	6	L2	CO2
	c.	Construct parallel binary adder/subtractor using full adder block and EX-OR gates. Also explain the operation of it.	6	L3	CO2
	d.	Design two-bit comparator using cascade connection of one-bit comparators and explain its operation.	4	L4	CO2

Module – 3

Q.5	a.	State transparency property in latches. What is the need for master-slave flip flops?	4	L1	CO3
	b.	With neat block diagram and truth-table explain the operation of master-slave JK flipflop.	6	L2	CO3
	c.	Design a synchronous mod-6 counter using JK flipflops.	6	L4	CO3
	d.	Implement mod-4-ring counter using shift registers.	4	L3	CO3

OR

Q.6	a.	Define register and shift register. Mention two applications of shift registers.	4	L1	CO3
	b.	With logic diagram and timing diagram explain the operation of positive edge triggered D-flip flop.	6	L2	CO3
	c.	Design a four-bit binary ripple up-counter with logic diagram and counting sequence and briefly explain its operation.	6	L4	CO3
	d.	Implement Mod-8 twisted ring counter using shift registers and write the count sequence.	4	L3	CO3

Module – 4

Q.7	a.	List the different relational operators available in verilog language.	4	L1	CO4
	b.	Explain different verilog data types with examples.	6	L2	CO4
	c.	For the circuit diagram shown in Fig.Q.7(c), develop a verilog program for the output Y in: i) data flow description ii) behavioral description.	6	L3	CO4

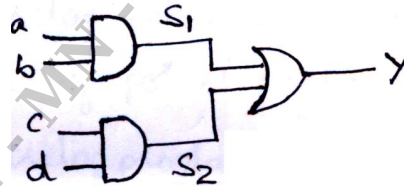


Fig.Q.7(c)

	d.	Develop a verilog program to implement 2×1 multiplexer using conditional operator. Also write the truth table of 2×1 multiplexer.	4	L4	CO4
OR					
Q.8	a.	List the different styles of descriptions in verilog programming.	4	L1	CO4
	b.	Explain verilog shift operators and arithmetic shift operators with examples.	6	L2	CO4
	c.	Let $A = 5'b11011$, $B = 5'b10101$, $C = 4'd3$. Determine the output of the following verilog program statements: i) $d = \&A$ ii) $e = \sim^4 b1011$ iii) $f = \sim(A \& (\sim B))$ iv) $g = A \parallel B$ v) $b = 3 ** 2$ vi) $i = \{2\{A\}\}$.	6	L3	CO4
	d.	Develop a verilog program for half subtractor using data flow description style by providing truth table and expressions.	4	L4	CO4
Module – 5					
Q.9	a.	Write the verilog format of if-else statement and explain it.	4	L1	CO4
	b.	Explain the operation of positive triggered JK flipflop by writing verilog code using case statement and truth table.	6	L2	CO4
	c.	Develop a verilog behavioral description code for calculating the factorial of positive integers.	6	L3	CO4
	d.	Develop a verilog program for D-latch using behavioral description style by providing truth table.	4	L4	CO4
OR					
Q.10	a.	Write the verilog format of case statement and explain it.	4	L1	CO4
	b.	Explain the operation of 2-to-1-line multiplexer by writing verilog structural description program and block diagram.	6	L2	CO4
	c.	Develop a verilog behavioral description code for three-bit binary up counter.	6	L3	CO4
	d.	Develop a verilog program for half adder using structural description style by providing truth table and expressions.	4	L4	CO4

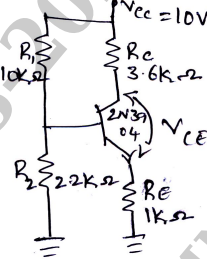
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Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Electronic Principles and Circuits

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With a neat circuit diagram, explain the voltage divider biasing circuit and also derive the expression.	10	L3	CO1
	b.	What is the collector-emitter voltage in Fig.Q1(b)	10	L3	CO1
 Fig.Q1(b)					
OR					
Q.2	a.	With diagram explain the two transistors model. Also derive $Z_{in}(\text{base})$.	10	L3	CO1
	b.	Explain the base biased amplifier circuit. Also explain AC equivalent circuit.	10	L3	CO1
Module – 2					
Q.3	a.	With diagram explain the enhancement model MOSFET. Draw Drain and Transconductance curve.	10	L3	CO2
	b.	Derive an expression of $i_D - V_{DS}$ relationship of NMOS transistor.	10	L3	CO2
OR					
Q.4	a.	Derive an expression of DC bias point and voltage gain of small signal operation of MOSFET.	10	L3	CO2
	b.	With a neat diagram explain the MOSFET T-equivalent circuit.	10	L3	CO2
Module – 3					
Q.5	a.	With diagram explain the R-2R ADC converter derive V_{out} .	10	L3	CO3
	b.	Derive V_{ref} and f_c of comparators with non zero reference to linear Amplifier.	10	L3	CO3
OR					
Q.6	a.	With neat diagram explain the operational amplifier base wein bridge oscillator circuit.	10	L3	CO3
	b.	Explain the operation of RC phase shift oscillator.	10	L3	CO3
Module – 4					
Q.7	a.	Briefly explain the four types of negative feedback.	10	L3	CO4
	b.	With diagram explain the ICVS amplifier circuit.	10	L3	CO4
OR					
Q.8	a.	With diagram explain the passband and stopband attenuation.	10	L3	CO4
	b.	Explain with circuit diagram of VCVS High pass filter.	10	L3	CO4
Module – 5					
Q.9	a.	With neat diagram explain the DC and AC two load line of VDB amplifier.	10	L3	CO5
	b.	Derive an expression of A_p of Class A power amplifier.	10	L3	CO5
OR					
Q.10	a.	With circuit and waveform explain the 1- ϕ RC triggering circuit.	10	L3	CO5
	b.	With neat diagram explain the Triac – Diac based bidirectional phase control circuit using SCR.	10	L3	CO5

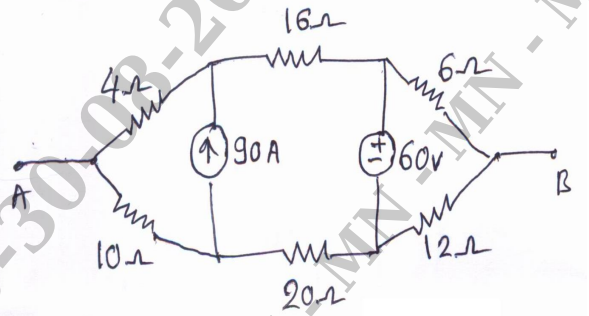
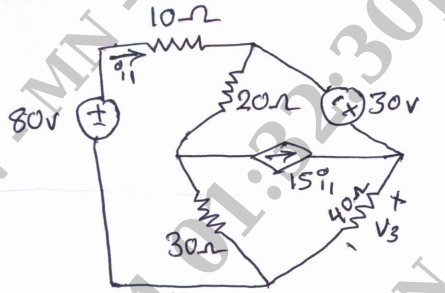
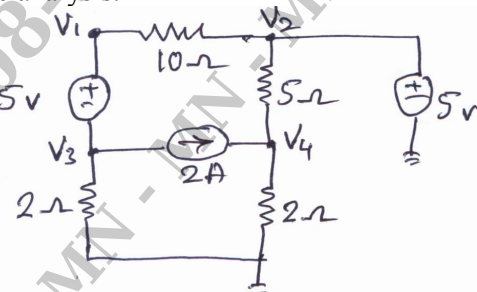
Third Semester B.E./B.Tech. Degree Examination, June/July 2024

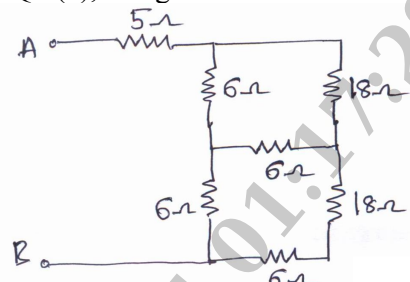
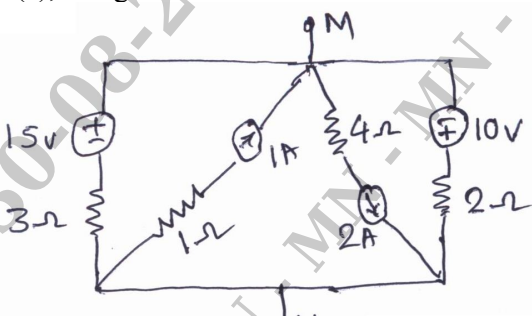
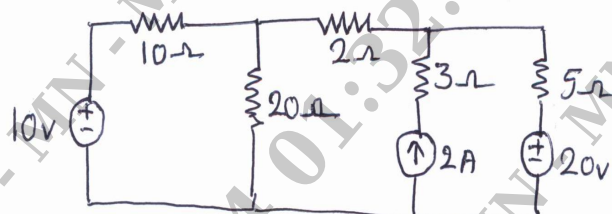
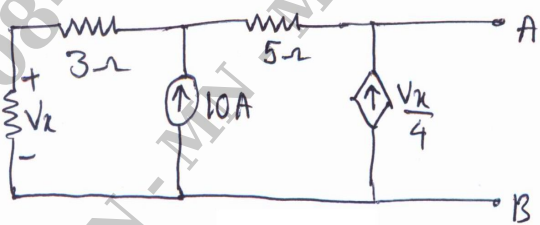
Network Analysis

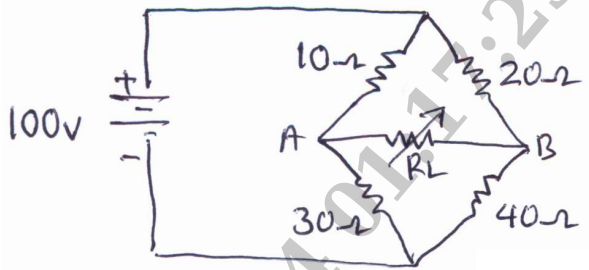
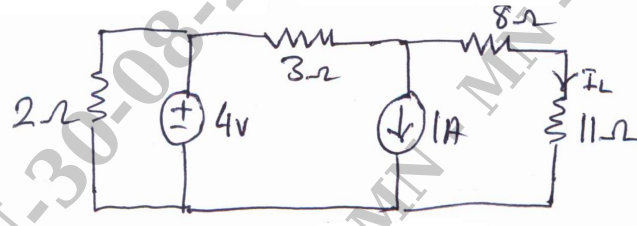
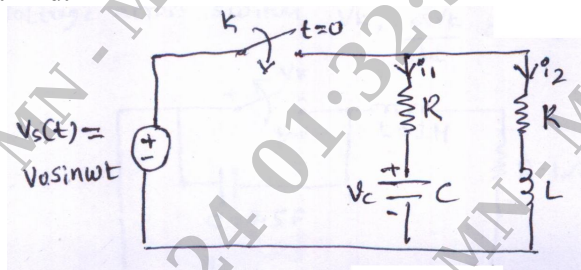
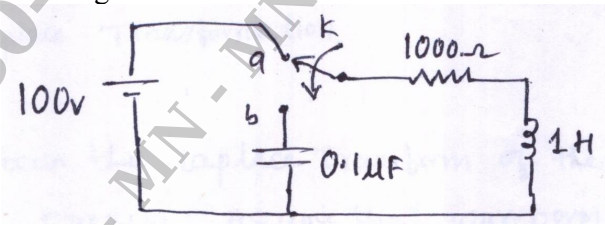
Time: 3 hrs.

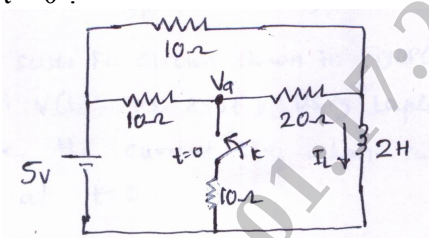
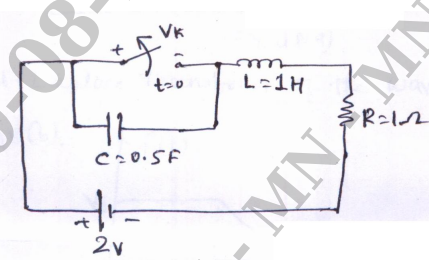
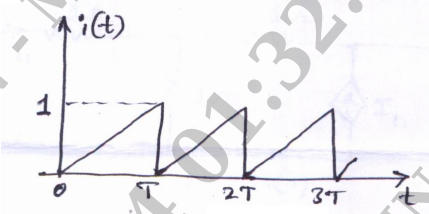
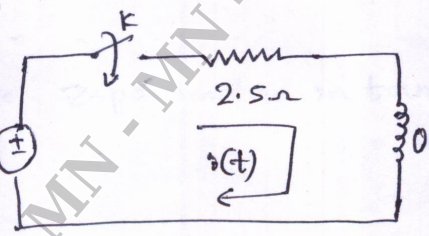
Max. Marks: 100

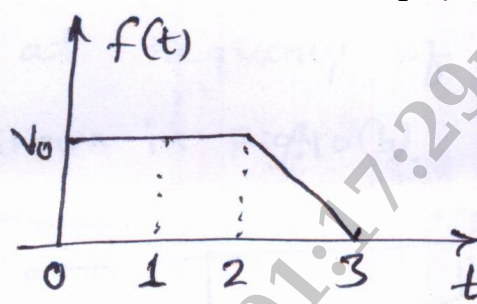
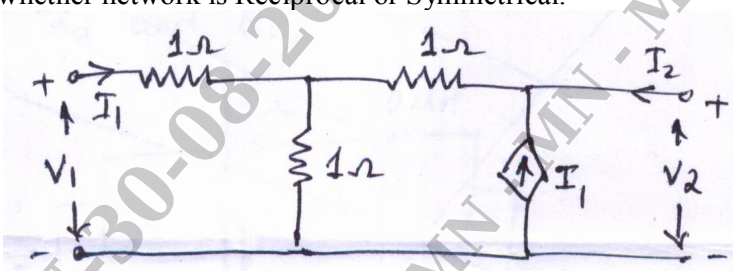
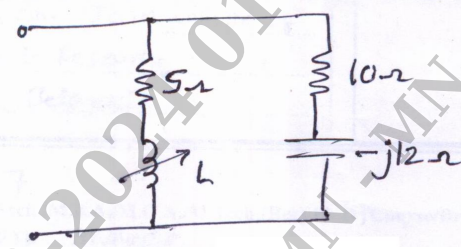
*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Reduce the network shown in Fig. Q1 (a) to a single voltage source in series with resistance between terminals A and B. Use source transformation and source shifting technique.		10	L3	CO1
	b.	Determine voltage V_3 in the circuit shown in Fig. Q1 (b), using loop analysis.		10	L3	CO1
OR						
Q.2	a.	For the network shown in Fig. Q2 (a), compute all node voltages V_1 , V_2 , V_3 and V_4 using Node analysis.		8	L3	CO1
			Fig. Q2 (a)			

	b. Determine the equivalent resistance between terminal A and B, in the network shown in Fig. Q2 (b), using star Delta transformation.  Fig. Q2 (b)	7	L3	CO1
	c. Find the potential difference between terminals M and N in the network shown in Fig. Q2 (c), using source transformation.  Fig. Q2 (c)	5	L3	CO1
Module – 2				
Q.3	a. Determine the voltage across 2Ω resistor in the circuit shown in Fig. Q3 (a), using the super position theorem.  Fig. Q3 (a)	10	L3	CO2
	b. Find Thevenin's equivalent at terminal A and B, in the network shown in Fig. Q3 (b).  Fig. Q3 (b)	10	L3	CO2

OR					
Q.4	a.	Determine the load resistance to receive maximum power from the source. Also find the maximum power delivered to the load in the circuit shown in Fig. Q4 (a).	8	L3	CO2
		 Fig. Q4 (a)			
	b.	For the circuit shown in Fig. Q4 (b), determine current I_L using Norton's theorem.	8	L3	CO2
		 Fig. Q4 (b)			
	c.	State Millman's theorem.	4	L2	CO2
Module – 3					
Q.5	a.	In the network shown in Fig. Q5 (a), a switch K is closed at $t = 0$. Determine $\frac{di_1}{dt}$, $\frac{di_2}{dt}$ at $t = 0^+$.	10	L3	CO3
		 Fig. Q5 (a)			
	b.	In the Network shown in Fig. Q5 (b), the switch K is changed position from a to b at $t = 0$. Solve for i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t = 0^+$. The circuit is reached steady state before switching.	10	L3	CO3
		 Fig. Q5 (b)			

OR					
Q.6	a.	In the network shown in Fig. Q6 (a), steady state has been reached with switch K open. At time $t = 0$, the switch is closed. Determine the value of $V_a(0^-)$ and $V_a(0^+)$ at $t = 0^+$.	10	L3	CO3
		 Fig. Q6 (a)			
	b.	In the network shown in Fig. Q6 (b), a steady state is reached with switch K closed. At $t = 0$, switch is opened. Determine voltage across switch V_K , $\frac{dV_K}{dt}$ at $t = 0^+$.	10	L3	CO3
		 Fig. Q6 (b)			
Module – 4					
Q.7	a.	State and prove initial and final value theorem in Laplace transformation.	10	L3	CO3
	b.	Obtain the Laplace transform of the waveform shown in Fig. Q7 (b). Assume that waveform is periodic.	10	L3	CO3
		 Fig. Q7 (b)			
OR					
Q.8	a.	In the series RL circuit shown in Fig. Q8 (a), the source voltage is $V(t) = 50 \sin 250t$ V. Using Laplace transform determine the current $i(t)$ when switch K is closed at $t = 0$.	10	L3	CO3
		 Fig. Q8 (a)			

	b.	Find Laplace transform of the waveform shown in Fig. Q8 (b).	10	L3	CO3
 Fig. Q8 (b)					
Module – 5					
Q.9	a.	Find Z and ABCD parameters for the network shown in Fig. Q9 (a). Also verify whether network is Reciprocal or Symmetrical.	10	L3	CO4
 Fig. Q9 (a)					
	b.	A series RLC circuit has a resistance of 10 Ω , an inductance of 0.3 H and a capacitance of 100 μ F. The applied voltage is 230 V. Find Resonance frequency, lower and upper cut-off frequencies, current at resonance, current at f_1 and f_2 , voltage across inductance at resonance.	10	L3	CO4
OR					
Q.10	a.	Derive Z-parameters in terms of H parameter.	8	L3	CO4
	b.	Find the value of L for which the circuit resonates at frequency of 1000 rad/sec, for the circuit shown in Fig. Q10 (b).	7	L3	CO4
 Fig. Q10 (b)					
	c.	Derive the relation between resonating frequency and half power frequencies i.e. $f_r = \sqrt{f_1 f_2}$	5	L2	CO4

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024

Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With a neat diagram, describe the functional units of a computer. Give examples for I/O.	10	L2	CO1
	b.	Write assembly language program for $X = (A * B) + (C * D)$ using one address, two address, and three address instructions formats.	06	L3	CO1
	c.	Explain the Bus structures.	04	L2	CO1
OR					
Q.2	a.	With a neat diagram, discuss the operational concepts in a computer highlighting the role of PC, MAR, MDR, IR.	10	L2	CO1
	b.	Discuss IEEE standard for single precision and double precision floating point numbers with standard notations.	06	L3	CO1
	c.	Distinguish between Big-endian and Little-endian memory assignment. With a neat sketch, show how the value 26789435 is stored using these methods.	04	L3	CO1
Module – 2					
Q.3	a.	Define addressing mode. Explain any five addressing mode with syntax and examples.	10	L2	CO2
	b.	What is subroutine? With a pseudocode or program segment illustrate parameter passing using register.	05	L2	CO2
	c.	Explain various assembler directives used in assembly language program.	05	L2	CO2
OR					
Q.4	a.	Explain stack operation with an example.	10	L2	CO2
	b.	Explain the shift and rotate operations with examples.	06	L2	CO2
	c.	Write a program to add 'n' number using indirect addressing mode.	04	L3	CO2
Module – 3					
Q.5	a.	Showing the possible registers configuration in I/O interface. Explain program controlled input/output.	10	L2	CO3
	b.	Explain in detail the situations where a number of devices capable of initiating interrupts are connected to processor. How to resolve the problems?	10	L2	CO3
OR					
Q.6	a.	What is an interrupt? With an example illustrate the concept of interrupt.	10	L2	CO3
	b.	Explain the Register involved in a DMA interface to illustrate DMA.	10	L2	CO3
Module – 4					
Q.7	a.	Illustrate internal structure of static memory.	10	L2	CO4
	b.	With a neat diagram, explain virtual memory organization.	10	L2	CO4
OR					
Q.8	a.	Classify memory in a computer. With a neat diagram, describe the organization of $2M \times 8$ DRAM chip.	10	L2	CO4
	b.	Briefly explain secondary storage devices.	06	L2	CO4
	c.	Explain use of a cache memory.	04	L2	CO4

Module – 5

Q.9	a.	List different ways of improving CPU performance. With a neat diagram, discuss three-bus organization of CPU.	10	L2	CO5
	b.	Discuss Hardwired control unit organization with relevant diagrams and illustrate the logic to generate Z_{in} control signal.	10	L3	CO5
OR					
Q.10	a.	Explain single-bus organization of data path in a processor with neat diagram, highlight the importance of gating signals.	10	L2	CO5
	b.	Develop the complete control signal sequence for the instruction $Add(R_1, R_3)$ with appropriate remarks.	06	L3	CO5
	c.	Discuss micro programmed control unit design with relevant diagrams.	04	L2	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Electromagnetics Theory

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	State and explain spherical coordinate system in detail.	5	L2	CO1
	b.	Four point charges each of 10 μC are placed in free space at the points (1, 0, 0), (-1, 0, 0), (0, 1, 0) and (0, -1, 0) m respectively. Determine the force on a point charge of 30 μC located at a point (0, 0, 1) m.	8	L3	CO1
	c.	Show that electric field intensity at a point, due to 'n' number of point charges, is given by, $E = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i}{R_i^2} a_{R_i} \text{ V/m}$	7	L3	CO1
OR					
Q.2	a.	Define electric field intensity. Derive the expression for electric field intensity due to infinite line charge.	9	L1	CO1
	b.	Given the two points A($\rho = 4.4, \phi = -115^\circ, Z = 2$) and B($x = -3.1, y = 2.6, z = -3$), find (i) The rectangular coordinate of point A (ii) The cylindrical coordinate of point B (iii) The distance between A and B.	5	L3	CO1
	c.	Find E at P(1, 5, 2) m in free space if a point charge of 6 μC is located at (0, 0, 1), the uniform line charge density $\rho_L = 180 \text{ nC/m}$ along x axis.	6	L3	CO1
Module – 2					
Q.3	a.	State and prove Gauss's law for point charge.	6	L3	CO2
	b.	Calculate the divergence of D at the point specified if, (i) $D = (2xyz - y^2)a_x + (x^2z - 2xy)a_y + x^2y a_z \text{ C/m}^2$ at $P_A(2, 3, -1)$ (ii) $D = 2\rho Z^2 \sin^2 \phi a_\rho + \rho Z^2 \sin 2\phi a_\phi + 2\rho^2 Z \sin^2 \phi a_z \text{ C/m}^2$ at $P_B(\rho = 2, \phi = 110^\circ, Z = -1)$ (iii) $D = 2r \sin \theta \cos \phi a_r + r \cos \theta \cos \phi a_\theta - r \sin \phi a_\phi \text{ C/m}^2$ at $P_C(r = 1.5, \theta = 30^\circ, \phi = 50^\circ)$	9	L3	CO2
	c.	Find electric field intensity at the point A(1, 2, -1) given the potential $V = 3x^2y + 2y^2z + 3xyz$	5	L3	CO2
OR					
Q.4	a.	Evaluate both sides of divergence theorem if $D = \frac{5r^2}{4} a_r \text{ C/m}^2$ in spherical co-ordinate for the volume enclosed by $r = 4 \text{ m}$ and $\theta = \frac{\pi}{4}$ radians.	8	L3	CO2

	b.	Calculate the work done in moving a charge 4C from B(1, 0, 0) to A(0, 2, 0) along the path $y = 2 - zx$, $z = 0$ in the field (i) $E = 5a_x$ V/m (ii) $E = 5xa_x$ V/m (iii) $E = 5xa_x + 5ya_y$ V/m	6	L3	CO2
	c.	Electrical potential at an arbitrary point in free space is given as, $V = 2(x+1)^2(y+2)^2(z+3)^2$ volt at a point P(2, -1, 4). Find (i) V (ii) E (iii) $ E $ (iv) $ D $ (v) ρ_v	6	L3	CO2
Module – 3					
Q.5	a.	Evaluate the expression for capacitance of two uniformly charged parallel planes of infinite extent.	8	L2	CO3
	b.	Determine whether or not the potential equations satisfies Laplaces equation : (i) $V = 2x^2 - 4y^2 + z^2$ (ii) $V = \phi \cos \phi + z$ (iii) $V = r^2 \cos \phi + \theta$	5	L3	CO3
	c.	An assembly of two concentric spherical shell is considered. The inner spherical shell is at a distance of 0.1 m and is at a potential of 0 volts. The outer spherical shell is at a distance of 0.2 m and at a potential of 100 V. The medium between them is a free space. Find E and D using spherical co-ordinate system.	7	L3	CO3
OR					
Q.6	a.	State and explain Biot-Savarts law applicable to magnetic field.	6	L2	CO3
	b.	Evaluate both sides of the stokes theorem for the field, $H = 6xya_x - 3y^2a_y$ A/m and the rectangular path around the region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $Z = 0$. Let the positive direction of ds be a_z .	8	L3	CO3
	c.	Let $A = (3y - z)a_x + 2xza_y$ wb/m in a certain region of free space. (i) Show that $\nabla \cdot A = 0$ (ii) At P(2, -1, 3) find A, B, H and J.	6	L3	CO3
Module – 4					
Q.7	a.	Obtain the expression for magnetic force between differential current elements.	6	L1	CO4
	b.	The point charge $Q = 18nC$ has a velocity of 5×10^6 m/s, in the direction $a_v = 0.60a_x + 0.75a_y + 0.30a_z$. Calculate the magnitude of force exerted on the charge by the field. (i) $B = -3a_x + 4a_y + 6a_z$ mT (ii) $E = -3a_x + 4a_y + 6a_z$ KV/m	6	L3	CO4
	c.	The magnetization in a magnetic material for which $\chi_m = 8$ is given in a certain region as $150 Z^2 a_x$ A/m. At $Z = 4$ cm, find the magnitude of, i) J_T ii) J iii) J_b .	8	L3	CO4
OR					
Q.8	a.	Obtain the magnetic boundary conditions at interface between two different magnetic material.	8	L2	CO4
	b.	Two differential current elements $I_1 dl_1 = 10^{-4} a_z$ Am at $P_1(1, 0, 0)$ and $I_2 dl_2 = 3 \times 10^{-6} (-0.5a_x + 0.4a_y + 0.3a_z)$ Am at $P_2(2, 2, 2)$ are located in free space. Find the vector force exerted on, (i) $I_2 dl_2$ by $I_1 dl_1$ (ii) $I_1 dl_1$ by $I_2 dl_2$	6	L3	CO4

	c.	The interface between two different regions is normal to one of three Cartesian axes. If $B_1 = \mu_0(43.5a_x + 24.0a_z)$ and $B_2 = \mu_0(22a_x + 24a_z)$. What is the ratio $\frac{\tan \theta_1}{\tan \theta_2}$?	6	L3	CO4
Module – 5					
Q.9	a.	For the given medium $\epsilon = 4 \times 10^{-9}$ F/m and $\sigma = 0$. Find K so that the following pair of fields satisfies Maxwell's equation, $E = (20y - Kt)a_x$ V/m, $H = (y + 2 \times 10^6 t)a_z$ A/m.	6	L3	CO5
	b.	Within a certain region $\epsilon = 10^{-11}$ F/m and $\mu = 10^{-5}$ H/m, If $B = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y$ T ; (i) Find E (ii) Find total magnetic flux passing through the surface $x = 0$, $0 < y < 40$ m, $0 < z < 2$ m at $t = 1$ μ sec.	8	L3	CO5
	c.	State and explain pointing theorem.	6	L2	CO5
OR					
Q.10	a.	Derive the modified Ampere's law by Maxwells for time varying fields.	5	L2	CO5
	b.	Show that the intrinsic impedance of the perfect dielectric $\eta = \frac{ E }{ H } = \sqrt{\frac{\mu}{\epsilon}}$ and show that its value in free space is 377Ω	7	L2	CO5
	c.	A plane electromagnetic wave having a frequency of 10 MHz has an average pointing vector of 1 W/m^2 . If medium is lossless with relative permeability of 2 and relative permittivity of 3 find (i) The velocity of propagation. (ii) Wavelength. (iii) Impedance of the medium (iv) rms electric field.	8	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is conditional probability? Prove that $P(B/A) = P(A/B) \cdot P(B) / P(A)$	05	L2	CO1
	b.	Define the autocorrelation and cross correlation. Discuss the properties of autocorrelation.	10	L2	CO1
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	05	L3	CO1
OR					
Q.2	a.	Define auto-covariance, random variable, cumulative distribution function and probability distribution function.	08	L1	CO1
	b.	The random variable its plot is given as $f_x(x) = 2 \cdot e^{-2x}$ for $x \geq 0$. Find the probability that it will take value between 1 and 3.	04	L3	CO1
	c.	Define probability with an example. Discuss their properties (axioms).	08	L2	CO1
Module – 2					
Q.3	a.	Explain amplitude modulation with necessary equations and sketches in time domain and frequency domain.	08	L3	CO2
	b.	Define modulation index and percentage of modulation. Explain over modulation and distortion.	06	L2	CO2
	c.	Derive the expression for Amplitude Modulation (AM) power in terms of modulation index.	06	L2	CO1
OR					
Q.4	a.	Explain a general block diagram of a frequency division multiplexing.	06	L1	CO2
	b.	Explain the working principle of lattice type balanced modulator with circuit diagram.	07	L1	CO2
	c.	With neat diagrams, explain high level collector modulator.	07	L2	CO2
Module – 3					
Q.5	a.	With a neat block diagram, explain converting a phase modulated signal into a frequency modulated signal.	07	L1	CO3
	b.	Determine the frequency modulated signal $v_{FM} = V_c \sin(2\pi f_c t + m_f \sin 2\pi f_m t)$ in terms of Bessel functions. Write the amplitude of sideband frequencies (J_n) in terms of modulation index (m_f).	06	L3	CO3
	c.	Identify the noise suppression of frequency modulated signal.	07	L2	CO3
OR					
Q.6	a.	What is the maximum bandwidth of an FM signal with a deviation of 30 kHz and a maximum modulating signal of 5 kHz. (i) Using number of sidebands $N = 9$ (ii) Using Carson's rule	04	L2	CO3
	b.	Define phase locked loop. Explain with neat circuit diagram of FM demodulator using the IC 565.	08	L2	CO3
	c.	With neat block diagram, explain the concept of frequency modulation with an IC voltage controlled oscillator (IC NE566)	08	L2	CO3

Module – 4

Q.7	a.	Why digitize the analog signals? Explain the different processes used to convert the analog signal to digital signal.	06	L2	CO4
	b.	What is quantization process? Explain the different types of quantization with their important characteristics.	07	L2	CO4
	c.	Explain the concept of Time division multiplexing with a neat block diagram.	07	L2	CO4

OR

Q.8	a.	Define PCM (Pulse Code Modulation). Explain the basic elements of a PCM system with neat diagrams.	06	L2	CO4
	b.	For the data stream 01101001. Draw the following line code waveforms: (i) Unipolar NRZ (ii) Polar NRZ (iii) Unipolar RZ (iv) Bipolar RZ (v) Manchester code (vi) Differential coding	09	L3	CO4
	c.	State and prove the sampling theorem. Explain with neat sketches and equations.	05	L2	CO4

Module – 5

Q.9	a.	Develop a code to generate and plot eye diagram.	06	L3	CO5
	b.	Define noise factor and noise figure. Also explain noise in cascade connection.	06	L2	CO5
	c.	Define Inter Symbol Interference (ISI). Outline baseband binary data transmission system with neat block diagram and equations.	08	L1	CO5

OR

Q.10	a.	Explain bandwidth requirements of TI systems.	06	L1	CO5
	b.	Write short notes on: (i) Signal to noise ratio (ii) External noise (iii) Internal noise	08	L1	CO5
	c.	An RF amplifier has an S/N ratio of 8 at the input and an S/N ratio of 6 at the output. What are the noise factor, noise figure and noise temperature?	06	L3	CO5

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CBCS SCHEME

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BEC403

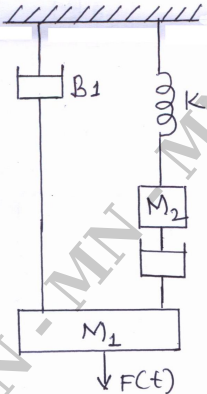
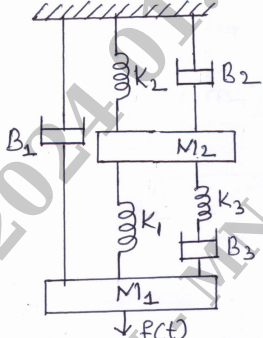
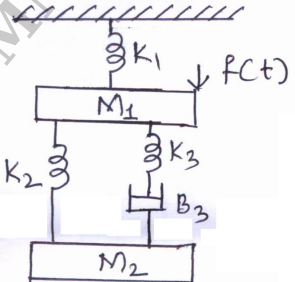
Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

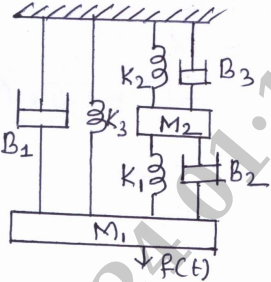
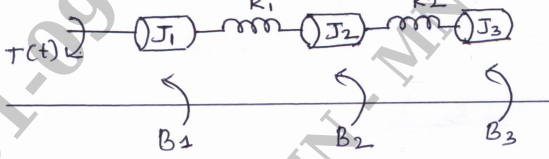
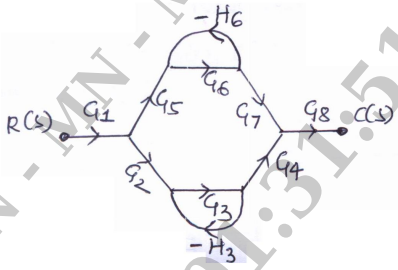
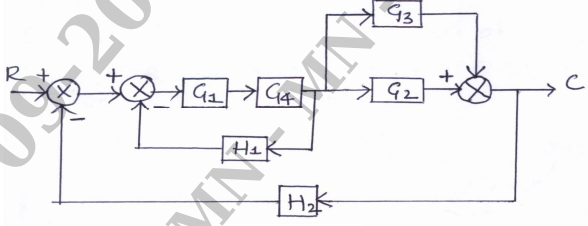
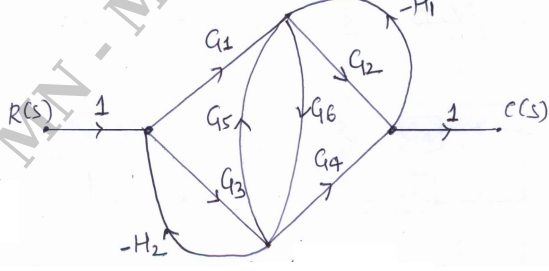
Control Systems

Time: 3 hrs.

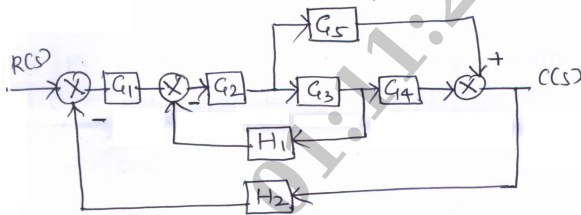
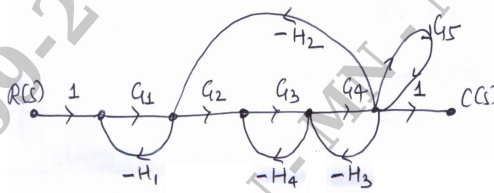
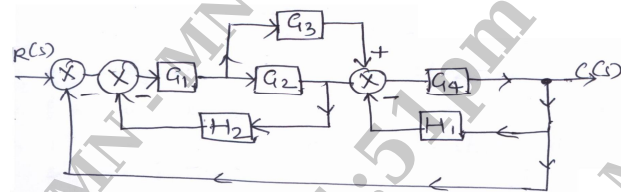
Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Define Control system. Write down any four differences between Open Loop Control System and Closed Loop Control System.		4	L2	CO1
	b.	For the mechanical system shown in Fig. Q1(b), obtain the equivalent electrical system using Force – Voltage method.	 Fig. Q1(b)	8	L2	CO1
	c.	For the mechanical system, shown in Fig. Q1(c), obtain the equivalent electrical system using Force – Current method.	 Fig. Q1(c)	8	L2	CO1
OR						
Q.2	a.	For the mechanical system shown in Fig. Q2(a), obtain the equivalent electrical system using Force – Voltage method.	 Fig. Q2(a)	7	L2	CO1

	b. For the mechanical system shown in Fig. Q2(b), obtain the equivalent electrical system using Force – Voltage method. Fig. Q2(b) 	7	L2	CO1
	c. Draw the electrical network based on torque – current analogy and write performance equation for the mechanical system of Fig. Q2(c). Fig. Q2(c) 	6	L2	CO1
Module – 2				
Q.3	a. Find $\frac{C(s)}{R(s)}$ by Mason's gain formula for Fig. Q3(a). Fig. Q3(a) 	6	L3	CO3
	b. Determine the transfer function $\frac{C(s)}{R(s)}$ of the system shown in Fig. Q3(b). Fig. Q3(b) 	6	L3	CO3
	c. For the single flow graph of Fig. Q3(c), find the transfer function using Mason's gain formula. Fig. Q3(c) 	8	L3	CO3

OR

Q.4	a.	Reduce the block diagram to its canonical form and obtain $C(s)/R(s)$ of the system of Fig. Q4(a).  Fig. Q4(a)	6	L3	CO3
	b.	Obtain the transfer function of the single flow graph shown in Fig. Q4(b), using Mason's gain formula.  Fig. Q4(b)	6	L3	CO3
	c.	Reduce the block diagram of Fig. Q4(c) to its simple form and obtain $C(s)/R(s)$.  Fig. Q4(c)	8	L3	CO3
Module – 3					
Q.5	a.	With the help of graphical representation and mathematical expression, explain the following test signals : i) Step signal ii) Ramp signal iii) Impulse signal iv) Parabolic signal.	8	L3	CO2
	b.	Find K_p, K_v, K_a and steady state error for a system with Open loop transfer function $G(s)H(s) = \frac{10(s+2)(s+3)}{s(s+1)(s+4)(s+5)}$, where $r(t) = 3 + t + t^2$.	6	L3	CO2
	c.	The Open loop transfer function of a servo system with unity feedback is given as $G(s) = \frac{10}{s(0.1s+1)}$. Find out static error constants and obtain steady state error when an input $r(t) = A_0 + A_1t + \frac{A_2}{2}t^2$ is applied.	6	L3	CO2
OR					
Q.6	a.	For a unity feedback control system with $G(s) = \frac{64}{s(s+9.6)}$, write the output response to a unit step input. Determine 1) The response at $t = 0.1$ set 2) Maximum value of response and the time at which it occurs. 3) Settling time.	10	L2	CO3

	b.	For the system shown in Fig. Q6(b), 1) Identify the type of $C(s) / E(s)$ 2) Find values of K_p , K_v , K_a . 3) If $r(t) = 10u(t)$, find steady state value of the output.	10	L2	CO3
		Fig. Q6(b)			
Module – 4					
Q.7	a.	Find the number of roots with positive real part, zero real part and negative real part for a system $s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$.	6	L2	CO4
	b.	For a unity feedback system , $G(s) = \frac{K}{s(1+0.4s)(1+0.25s)}$, find range of values of K, Marginal value of K and frequency of sustained oscillations.	6	L2	CO4
	c.	Explain the angle condition in Root locus. Test the following points using angle condition for the system $G(s) H(s) = \frac{K}{s(s+2)(s+4)}$ i) $s = -0.75$ ii) $s = -1 + j4$.	8	L2	CO4
OR					
Q.8	a.	Sketch the complete root locus and comment on the stability of the system $G(s) H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$.	12	L2	CO4
	b.	Sketch the Bode plot for the transfer fl. Find value of 'K' for $W_{gc} = 5$ rad/sec. $G(s) = \frac{K s^2}{(1+0.2s)(1+0.02s)}$	8	L2	CO4
Module – 5					
Q.9	a.	For a certain control system $G(s) H(s) = \frac{K}{s(s+2)(s+10)}$, sketch the Nyquist plot and hence calculate the range values of K for stability.	10	L2	CO5
	b.	Explain the Lag compensator and Lead compensator with the help of a circuit diagram.	10	L2	CO5
OR					

Q.10	a.	Construct the state model using phase variables if the system is described by the differential equation $\frac{d^3y(t)}{dt^3} + 4\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 2y(t) = 5u(t)$. Also draw the state diagram.	6	L2	CO5
	b.	The transfer function of a control system is $\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 4}{s^3 + 2s^2 + 3s + 2}$. Obtain the State model using signal flow graph.	7	L2	CO5
	c.	Find the state transition matrix for $A = \begin{bmatrix} 0 & -1 \\ +2 & -3 \end{bmatrix}$	7	L1	CO5

CBCS SCHEME

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BEC405A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Bring out the difference between Microprocessor and Microcontroller.	6	L2	CO1
	b.	With a neat Architecture diagram, explain the Architecture of 8051 Microcontroller.	10	L2	CO1
	c.	Explain : (i) RST (ii) INT Pins of 8051	4	L1	CO1
OR					
Q.2	a.	Differentiate between CISC and RISC.	6	L2	CO1
	b.	With a neat diagram, explain the Internal Memory Structure and Programming Model of 8051 Microcontroller.	10	L2	CO1
	c.	List out special features of 8051 Microcontroller.	4	L2	CO1
Module – 2					
Q.3	a.	Define Addressing Mode. Explain different addressing modes with example.	10	L2	CO2
	b.	Write an ALP to add two 16-bit numbers loaded in R ₁ R ₀ and R ₃ R ₂ . Store the result in R ₆ R ₅ and R ₄ from MSB to LSB.	10	L3	CO2
OR					
Q.4	a.	Define Stack. Explain the operation of Stack using Stack Pointer, PUSH and POP Instructions.	10	L2	CO2
	b.	Write an ALP to find largest of N numbers.	10	L3	CO2
Module – 3					
Q.5	a.	Explain : (i) TMOD (ii) TCON register of 8051.	10	L2	CO3
	b.	Assume XTAL = 22 MHz. Write an ALP to generate a square of frequency 1 kHz on Pin P1.2.	10	L2	CO3
OR					
Q.6	a.	Explain : (i) SCON register (ii) Importance of TI Flag	10	L2	CO3
	b.	Write a C program to transfer “YES” serially at 9600 baud rate, 8 bit data, 1 stop bit, do this continuously.	10	L3	CO3
Module – 4					
Q.7	a.	Define Interrupt. List the steps involved in Executing an Interrupt.	10	L2	CO4
	b.	Explain Interrupt Vector table of 8051 Microcontroller.	5	L2	CO4
	c.	Explain Interrupt enable register.	5	L2	CO4
OR					
Q.8	a.	Explain Interrupt Control used in 8051.	10	L2	CO4
	b.	Explain the steps involved in programming serial communication Interrupt.	5	L2	CO4

	c.	Explain how multiple Interrupts are handled in 8051.	5	L2	CO4
Module – 5					
Q.9	a.	Explain DAC Interface with a neat diagram and also write a program to generate staircase waveform.	10	L3	CO5
	b.	With a neat diagram, write a program to Interface Stepper Motor to 8051 Microcontroller.	10	L3	CO5
OR					
Q.10	a.	Explain the Interfacing of DC motor using C programming.	10	L3	CO5
	b.	With a neat diagram, write a ALP to Interface LCD to 8051 Microcontroller.	10	L3	CO5

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024
AV Mathematics – III for EC/BM Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book and statistical table are permitted.

3. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1				M	L	C															
Q.1	a.	Find the Fourier series for $f(x)=\begin{cases} -K, & \text{in } (-\pi,0) \\ K & \text{in } (0,\pi) \end{cases}$ and hence deduce $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+.....$	6	L2	CO1																
	b.	Expand $f(x)=2x-1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																
	c.	Express y as a Fourier series upto the first harmonics given the following values: <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>4</td><td>8</td><td>15</td><td>7</td><td>6</td><td>2</td></tr></table>	x	0	1	2	3	4	5	y	4	8	15	7	6	2	7	L3	CO1		
x	0	1	2	3	4	5															
y	4	8	15	7	6	2															
OR																					
Q.2	a.	Find the Fourier series for $f(x)=x-x^2$ in $-1 < x < 1$.	6	L2	CO1																
	b.	Show that half range sine series of $f(x)=\pi x-x^2$ in the interval $(0,\pi)$ is $\frac{8}{\pi}\sum_{h=0}^{\infty}\frac{1}{(2n+1)^3}\sin(2n+1)x$	7	L2	CO1																
	c.	Obtain the Fourier series of y upto 2 nd harmonics f(x) is given by <table border="1"><tr><td>x</td><td>0</td><td>$\pi/3$</td><td>$2\pi/3$</td><td>π</td><td>$4\pi/3$</td><td>$5\pi/3$</td><td>2π</td></tr><tr><td>f(x)</td><td>1.98</td><td>1.30</td><td>1.05</td><td>1.30</td><td>-0.88</td><td>-0.25</td><td>1.98</td></tr></table>	x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π	f(x)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98	7	L3	CO1
x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π														
f(x)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98														
Module – 2																					
Q.3	a.	Find the Fourier transform of $f(x)=\begin{cases} 1-x^2, & x <1 \\ 0 & x \geq1 \end{cases}$ and hence find the value of $\int_0^{\infty}\frac{x\cos x-\sin x}{x^3}dx$	6	L2	CO2																
	b.	Find the Fourier sine and cosine transform of $f(x)=e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																
	c.	Solve the integral equation $\int_0^{\infty}f(\theta)\cos\alpha\theta d\theta=\begin{cases} 1-\alpha, & 0\leq\alpha\leq1 \\ 0, & \alpha>1 \end{cases}$ and hence evaluate $\int_0^{\infty}\frac{\sin^2t}{t^2}dt$.	7	L3	CO2																

1 of 3

OR

Q.4	a.	Find the Fourier transform of $e^{-a^2x^2}$, $a > 0$.	6	L2	CO2
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$, $m > 0$	7	L2	CO2
	c.	Find the discrete Fourier transform of the sequence $\{1, 2, 1, 3\}^T$.	7	L3	CO2

Module – 3

Q.5	a.	Obtain the Z-transform i) $\cos n\theta$ ii) $\sin n\theta$.	6	L2	CO3
	b.	Find the inverse Z-transform of $\frac{3z^2 + 2z}{(5z-1)(5z+2)}$	7	L2	CO3
	c.	Solve by using Z-transforms : $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$.	7	L3	CO3

OR

Q.6	a.	Find the Z-transform of $2n + \sin\left(\frac{n\pi}{4}\right) + 1$	6	L2	CO3
	b.	Find the inverse Z-transform of $\frac{4z^2 - 2z}{(z-1)(z-2)^2}$.	7	L2	CO3
	c.	If $\bar{u}(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$ find the value of u_0, u_1, u_2 .	7	L3	CO3

Module – 4

Q.7	a.	Solve $(D^4 + 8D^2 + 16)y = 0$.	6	L1	CO4
	b.	Solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 13y = e^{3t} \cosh 2t$.	7	L2	CO4
	c.	Solve $x^3 + x^2y'' + xy' + 8y = 65 \cos(\log x)$.	7	L3	CO4

OR

Q.8	a.	Solve $y'' + 9y = \cos 2x \cos x$.	6	L2	CO4
	b.	Solve $(2x+1)^2 y'' - 2(2x+1)y' - 12y = 6x + 5$.	7	L2	CO4
	c.	In an LCR circuit, the charge q on a plate of a condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{c} = E \sin pt$. Solve the equation for q .	7	L3	CO4

Module – 5

Q.9	a.	Fit a straight line for the following data: <table><tr><td>x</td><td>50</td><td>70</td><td>100</td><td>120</td></tr><tr><td>y</td><td>12</td><td>15</td><td>21</td><td>25</td></tr></table>	x	50	70	100	120	y	12	15	21	25	6	L1	CO5												
x	50	70	100	120																							
y	12	15	21	25																							
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>9</td><td>8</td><td>10</td><td>12</td><td>11</td><td>13</td><td>14</td></tr></table>	x	1	2	3	4	5	6	7	y	9	8	10	12	11	13	14	7	L2	CO5						
x	1	2	3	4	5	6	7																				
y	9	8	10	12	11	13	14																				
	c.	Compute the rank correlation coefficient for the following data: <table><tr><td>x</td><td>68</td><td>63</td><td>75</td><td>50</td><td>62</td><td>80</td><td>78</td><td>40</td><td>55</td><td>60</td></tr><tr><td>y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>	x	68	63	75	50	62	80	78	40	55	60	y	62	58	68	45	81	60	68	48	50	70	7	L3	CO5
x	68	63	75	50	62	80	78	40	55	60																	
y	62	58	68	45	81	60	68	48	50	70																	
OR																											
Q.10	a.	An experiment on life time 't' of cutting tool at different cutting speeds v(units) are given below <table><tr><td>Speed (v)</td><td>350</td><td>400</td><td>500</td><td>600</td></tr><tr><td>Life (t)</td><td>61</td><td>26</td><td>7</td><td>2.6</td></tr></table> Fit a relation of the form $v = at^b$.	Speed (v)	350	400	500	600	Life (t)	61	26	7	2.6	6	L2	CO5												
Speed (v)	350	400	500	600																							
Life (t)	61	26	7	2.6																							
	b.	The following data gives the age of husband (x) and the age of wife (y) in years. Form the 2 regression lines and calculate the age of husband corresponding to 16 years of age of wife. <table><tr><td>x</td><td>36</td><td>23</td><td>27</td><td>28</td><td>28</td><td>29</td><td>30</td><td>31</td><td>33</td><td>35</td></tr><tr><td>y</td><td>29</td><td>18</td><td>20</td><td>22</td><td>27</td><td>21</td><td>29</td><td>27</td><td>29</td><td>28</td></tr></table>	x	36	23	27	28	28	29	30	31	33	35	y	29	18	20	22	27	21	29	27	29	28	7	L2	CO5
x	36	23	27	28	28	29	30	31	33	35																	
y	29	18	20	22	27	21	29	27	29	28																	
	c.	If the coefficient of correlation between the variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}(3/5)$. Show that $\sigma_y = 2\sigma_x$.	7	L3	CO5																						

CBCS SCHEME

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Cell? Explain Prokaryotic cells and Eukaryotic cell in brief.	10	L2	CO1
	b.	With a neat figure, explain the structure of plant cell and animal cell.	10	L2	CO1
OR					
Q.2	a.	What is a biomolecule? Explain the classification of biomolecule.	10	L1	CO1
	b.	Explain the properties and functions of Enzymes.	10	L2	CO1
Module – 2					
Q.3	a.	What are Bioplastics? Compare the properties of PHA and PLA as bioplastics.	10	L1	CO1
	b.	With an example, explain the development of DNA vaccines.	10	L3	CO2
OR					
Q.4	a.	Explain the importance of lipids and its application in cleaning agents.	10	L2	CO1
	b.	Discuss the development of vaccine for Covid 19.	10	L3	CO2
Module – 3					
Q.5	a.	Discuss about the Human Brain as a CPU system.	10	L3	CO2
	b.	What is ECG? Describe the various parts of ECG.	10	L2	CO1
OR					
Q.6	a.	Describe the kidney as a filtration system.	10	L3	CO2
	b.	What are Pacemakers? Briefly explain the various kinds of pacemakers.	10	L2	CO2
Module – 4					
Q.7	a.	What is Echolocation? Discuss the application of echolocation in Ultrasonography.	10	L1	CO2
	b.	Explain the structure and design of Kingfisher's beak led to bullet trains.	10	L3	CO2
OR					
Q.8	a.	Discuss about the human blood substitutes.	10	L1	CO1
	b.	What is a lotus effect? Explain the mechanism and applications of lotus leaf effect.	10	L4	CO2

Module – 5					
Q.9	a.	Explain the architecture of Muscular and Skeletal systems as Scaffolds.	10	L3	CO4
	b.	Discuss the applications of Artificial intelligence in the diagnosis of disease.	10	L4	CO4
OR					
Q.10	a.	What is Bioprinting? Discuss the process and applications of Bio printing.	10	L4	CO4
	b.	Write a note on : (i) Electrical Tongue (ii) Self-healing Bioconcrete.	10	L3	CO4

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